

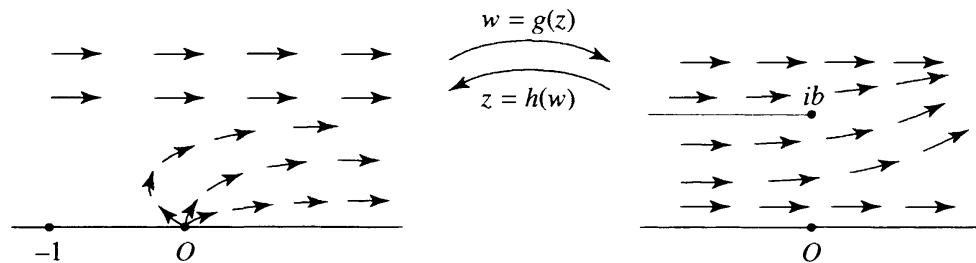
A and B are determined by the boundary conditions, with final result

$$g(z) = \frac{b}{\pi} (1 + z + \operatorname{Log} z), \quad \operatorname{Im} z > 0.$$

A flow function corresponding to a source in the tributary in the w -plane then corresponds to a flow function corresponding to a source at $z = 0$ in the z -plane. This is given by $\operatorname{Arg} z = \operatorname{Arg} h(w)$, where $h(w)$ is the inverse of the conformal map $g(z)$. We superimpose this upon a flow in the river, to obtain a flow function of the form

$$\psi(w) = c_1 \operatorname{Arg} h(w) + c_2 \operatorname{Im} h(w),$$

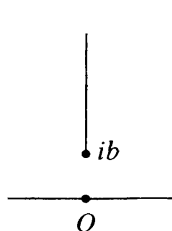
where the constants c_1 and c_2 are adjusted to the magnitudes of the flows.



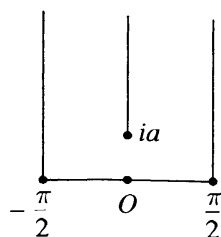
Exercises for XI.4

- Let D be the domain obtained from the upper half-plane by deleting the part of the imaginary axis above ib . Find a conformal map $w = g(z)$ of the upper half-plane onto D such that $g(0) = -\infty$, $g(\infty) = +\infty$, and $g(-1) = ib$. Solve for $z = h(w)$. With $\psi(w) = \arg(h(w))$ as a stream function, use the map to sketch the streamlines of the flow of water in a river as it passes under a vertical gate. *Hint.* In the Schwarz-Christoffel formula, remember to take the angles at ∞ to be negative. An alternative method is to use an inversion to map the infinite vertical slit onto a finite slit from 0 to ia . The answer is

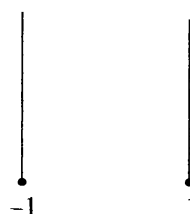
$$g(z) = \frac{b}{2} \left(\sqrt{z} - \frac{1}{\sqrt{z}} \right), \quad h(w) = 1 + 2\frac{w^2}{b^2} + 2\frac{w}{b} \sqrt{\frac{w^2}{b^2} + 1}.$$



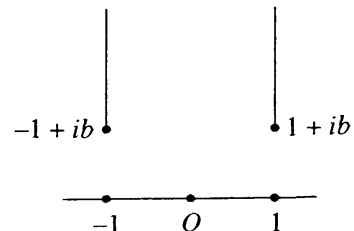
Exercise 1



Exercise 2



Exercise 3



Exercise 4

2. Let D be the domain obtained from the vertical half-strip $\{\operatorname{Im} z > 0, |\operatorname{Re} z| < \pi/2\}$ by deleting the part of the imaginary axis above ia . Find a conformal map $w = g(z)$ of the upper half-plane onto D . Solve for $z = h(w)$. Use the map to sketch the streamlines of the flow of water in a channel with a hairpin turn. *Hint.* Use the inverse sine function to map the half-strip onto an upper half-plane (see Exercise I.11), and refer to Exercise 1.
3. Find a conformal map $w = g(z)$ of the upper half-plane onto the plane with two vertical slits from ± 1 to $\pm 1 + i\infty$ deleted. Use the map to sketch roughly the flow of water near the mouth of a channel that empties into the middle of a large reservoir. Express the stream function for the flow in terms of the inverse $z = h(w)$ of $g(z)$.
4. Let D be the upper half-plane with two vertical slits from $\pm 1 + ib$ to $\pm 1 + i\infty$, and let $w = g(z)$ be the conformal map of the upper half-plane onto D satisfying $w(0) = 0$, $w(-1) = -\infty$, and $w(+1) = +\infty$. Show that the map is symmetric about the imaginary axis, so that $w(i\infty) = i\infty$. Express the map as an integral. Suppose in your integral formula that the point $a > 1$ is mapped to the end $1 + ib$ of the slit. Sketch the streamlines of the flow in the w -plane that corresponds to the stream function $\psi(z) = \arg(z - a) - \arg(z + a)$ in the z -plane. What are its sources and sinks?

5. Compactness of Families of Functions

We have reached a turning point. Our proof of the Riemann mapping theorem requires some technical ingredients at a higher level of mathematical sophistication than what we have used heretofore. In this section we summarize some of the ideas involved in the Arzelà-Ascoli theorem, and we use it to prove Montel's theorem (thesis grade). Montel's theorem is one of the key ingredients of the proof of the Riemann mapping theorem given in the next section. For an expanded treatment of the background material, see any good introduction to mathematical analysis.

Let E be a subset of the complex plane $\mathbb{C}$, and let $\mathcal{F}$ be a family of complex-valued functions on E . We say that $\mathcal{F}$ is **equicontinuous** at a point $z_0 \in E$ if for any $\varepsilon > 0$, there is $\delta > 0$ such that if $z \in E$ satisfies $|z - z_0| < \delta$, then $|f(z) - f(z_0)| < \varepsilon$ for all $f \in \mathcal{F}$. This coincides with the “ ε - δ ” definition of continuity at a point z_0 , except that the same δ must serve simultaneously for *all* functions f in the family $\mathcal{F}$.

We say that the family $\mathcal{F}$ is **uniformly bounded** on E if there is a constant $M > 0$ such that $|f(z)| \leq M$ for all $z \in E$ and all $f \in \mathcal{F}$.

One condition guaranteeing equicontinuity of a family of functions is that the derivatives of the functions in the family be uniformly bounded. Suppose, for instance, that $\mathcal{F}$ is a family of analytic functions on a domain D ,

and suppose there is $M > 0$ such that $|f'(z)| \leq M$ for all $z \in D$ and all $f \in \mathcal{F}$. Integrating along the straight line segment from z_0 to a nearby point z , we obtain

$$|f(z) - f(z_0)| = \left| \int_{z_0}^z f'(\zeta) d\zeta \right| \leq M|z - z_0|, \quad f \in \mathcal{F},$$

for z near z_0 . This estimate implies that the family $\mathcal{F}$ is equicontinuous at z_0 .

Recall that a subset E of the complex plane $\mathbb{C}$ is **compact** if it is closed and bounded. We are now prepared to state one version of the **Arzelà-Ascoli theorem**.

Theorem. *Let E be a compact subset of $\mathbb{C}$, and let $\mathcal{F}$ be a family of continuous complex-valued functions on E that is uniformly bounded. Then the following are equivalent.*

- (1) *The family $\mathcal{F}$ is equicontinuous at each point of E .*
- (2) *Each sequence of functions in $\mathcal{F}$ has a subsequence that converges uniformly on E .*

In the sequel we will use only the forward implication, that (1) implies (2). The proof that (1) implies (2) proceeds in outline as follows. Let $\{f_n\}$ be a sequence in $\mathcal{F}$. Let $\{z_j\}$ be a sequence of points in E that is dense in E . First one uses a standard diagonalization argument to find a subsequence $\{f_{n_k}\}$ such that $f_{n_k}(z_j)$ converges for each j . Then one uses the equicontinuity of the f_{n_k} 's and the density of the z_j 's to show that $\{f_{n_k}\}$ converges uniformly on E . The proof of the converse, that (2) implies (1), is an easy exercise in "proof by contradiction." For more details, see any good introduction to mathematical analysis.

The proof of the Arzelà-Ascoli theorem actually extends to a quite general situation. The theorem is valid for a family of functions on any compact metric space, and there is a version of the theorem for arbitrary compact topological spaces. The theorem also holds for functions whose ranges lie in any compact metric space. In particular, the theorem holds for functions from a compact set E to the extended complex plane $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$, provided that we use the spherical metric to measure distances in the range space. To be more precise, we denote by $\sigma(z, w)$ the spherical distance from z to w , as in Section IX.3. We say that a sequence of functions $\{f_n\}$ on E **converges uniformly to f in the spherical metric** if $\sigma(f_n(z), f_m(z))$ tends to 0 uniformly for $z \in E$ as $n, m \rightarrow \infty$. A family $\mathcal{F}$ is **equicontinuous with respect to the spherical metric** at $z_0 \in E$ if for any $\varepsilon > 0$ there is $\delta > 0$ such that if $z \in E$ satisfies $|z - z_0| < \delta$, then $\sigma(f(z), f(z_0)) < \varepsilon$ for all $f \in \mathcal{F}$. When we combine the Arzelà-Ascoli theorem with the diagonalization argument used above, we obtain the following version of the theorem, which we record for use when we return to this circle of ideas in Chapter XIV.

Theorem. *Let D be a domain in the complex plane, and let $\mathcal{F}$ be a family of continuous functions from D to the extended complex plane $\mathbb{C}^*$. Then the following are equivalent.*

- (1) *Any sequence in $\mathcal{F}$ has a subsequence that converges uniformly on compact subsets of D in the spherical metric.*
- (2) *The family $\mathcal{F}$ is equicontinuous at each point of D , with respect to the spherical metric.*

For now, our applications to complex analysis will be based on the following preliminary version of Montel's theorem, which was obtained by P. Montel in his thesis. We will prove a stronger version in Chapter XIV.

Theorem. *Suppose $\mathcal{F}$ is a family of analytic functions on a domain D such that $\mathcal{F}$ is uniformly bounded on each compact subset of D . Then every sequence in $\mathcal{F}$ has a subsequence that converges normally on D , that is, uniformly on each compact subset of D .*

If $z_0 \in D$, there is $r > 0$ such that the closed disk $\{|z - z_0| \leq r\}$ is contained in D . The family $\mathcal{F}$ is uniformly bounded on the disk. By the Cauchy estimates, the derivatives of the functions in $\mathcal{F}$ are uniformly bounded on the smaller disk $\{|z - z_0| \leq r - \varepsilon\}$. Hence $\mathcal{F}$ is equicontinuous at each $z_0 \in D$. To complete the proof we combine the Arzelà-Ascoli theorem with a diagonalization argument. Let E_n be the set of $z \in D$ such that $|z| \leq n$ and the distance from z to ∂D is at least $1/n$. Then E_n is compact, the E_n 's increase to D , and any compact subset of D is contained in some E_n . Let $\{f_n\}$ be a sequence in $\mathcal{F}$. By the Arzelà-Ascoli theorem, there is a subsequence $f_{11}, f_{12}, f_{13}, \dots$ that converges uniformly on E_1 . This has a further subsequence $f_{21}, f_{22}, f_{23}, \dots$ that converges uniformly on E_2 , and so on. The diagonal sequence $f_{11}, f_{22}, f_{33}, \dots$ is then a subsequence of the original sequence that converges uniformly on each E_n , hence uniformly on each compact subset of D .

One typical application of Montel's theorem is to guarantee the existence of extremal functions for "compact" extremal problems. To illustrate the principle, we consider the extremal problem of maximizing the derivative at a prescribed point over a compact family of analytic functions.

Let D be a domain, and fix a point $z_0 \in D$. Let $\mathcal{F}$ be the family of analytic functions $f(z)$ on D such that $|f(z)| \leq 1$ on D . The extremal problem is to maximize $|f'(z_0)|$ among all functions $f \in \mathcal{F}$. The **extremal value** for the problem is

$$A = \sup \{|f'(z_0)| : f \in \mathcal{F}\}.$$

Since the functions in $\mathcal{F}$ are uniformly bounded on D , their derivatives are uniformly bounded at z_0 , and A is finite. A function $G \in \mathcal{F}$ such that $|G'(z_0)| = A$ is an **extremal function** for the problem. The existence of an extremal function follows almost immediately from Montel's theorem.

If $\{f_n(z)\}$ is any sequence of functions in $\mathcal{F}$ such that $|f'_n(z_0)| \rightarrow A$, then by Montel's theorem, the f_n 's have a subsequence that converges normally on D to an analytic function $G(z)$. On account of the normal convergence, $|G(z)| \leq 1$ and $|G'(z_0)| = A$.

Theorem. *Let D be a domain in the complex plane on which there is a nonconstant bounded analytic function, and let $z_0 \in D$. Then there is an analytic function $G(z)$ on D such that $|G(z)| \leq 1$ for $z \in D$, and $|f'(z_0)| \leq |G'(z_0)|$ for any analytic function $f(z)$ on D satisfying $|f(z)| \leq 1$ on D . Further, $G(z_0) = 0$ and $G'(z_0) \neq 0$.*

It remains only to establish the final statement. Let $h(z)$ be a nonconstant bounded analytic function on D . Then $h(z) - h(z_0)$ has a zero of finite order, say of order N , at z_0 . If $\varepsilon > 0$ is tiny, the function $f(z) = \varepsilon(h(z) - h(z_0))/(z - z_0)^{N-1}$ satisfies $|f(z)| \leq 1$ and $f'(z_0) \neq 0$. Consequently, $A = |G'(z_0)| > 0$. Finally, $g(z) = (G(z) - G(z_0))/(1 - \overline{G(z_0)}z)$ is the composition of $G(z)$ and a conformal self-map of Δ , so that $g \in \mathcal{F}$. From

$$|G'(z_0)| \geq |g'(z_0)| = \frac{|G'(z_0)|}{1 - |G(z_0)|^2},$$

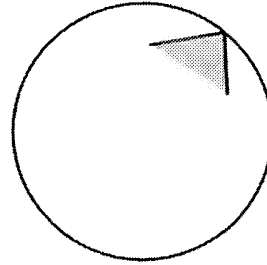
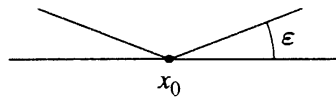
we deduce that $G(z_0) = 0$.

The extremal function $G(z)$ is called the **Ahlfors function** of D . The Ahlfors function depends on z_0 . However, it can be shown (Exercise 9) that the Ahlfors function corresponding to a fixed point $z_0 \in D$ is unique, up to multiplication by a unimodular constant. The extremal value $A = |G'(z_0)|$ can be regarded as the best constant for which the Schwarz lemma (infinitesimal version) holds with respect to $z_0 \in D$. We will show in the next section that when D is simply connected, the Ahlfors function maps D conformally onto the open unit disk.

Exercises for XI.5

1. Let $\{f_n(z)\}$ be a uniformly bounded sequence of analytic functions on a domain D , and let $z_0 \in D$. Suppose that for each $m \geq 0$, $f_n^{(m)}(z_0) \rightarrow 0$ as $n \rightarrow \infty$. Show that $f_n(z) \rightarrow 0$ normally on D .
2. Let $\{f_n(z)\}$ be a sequence of analytic functions on a domain D , and let $z_0 \in D$. Suppose $\operatorname{Re} f_n(z) \geq -C$ for all $z \in D$, and $f_n^{(m)}(z_0) \rightarrow 0$ as $n \rightarrow \infty$ for each $m \geq 0$. Show that $f_n(z) \rightarrow 0$ normally on D . *Hint.* Use Problem 1, which is a special case.
3. Let $f(z)$ be a bounded analytic function on the horizontal strip $\{-1 < \operatorname{Im} z < 1\}$, such that $f(x)$ tends to 0 as x tends to $+\infty$. Show that for any $\varepsilon > 0$, $f(x + iy)$ tends to 0 as x tends to $+\infty$, uniformly for $-1 + \varepsilon < y < 1 - \varepsilon$.

4. Let $f(z)$ be a bounded analytic function on the upper half-plane. (a) Show that if $f(z)$ tends to 0 as z tends to ∞ along the imaginary axis, then for any $\varepsilon > 0$, $f(z)$ tends to 0 uniformly as z tends to ∞ through the sector $\{\varepsilon < \arg z < \pi - \varepsilon\}$. (b) Show that if $f(z)$ tends to a limit L as $z = x_0 + iy$ tends to $x_0 \in \mathbb{R}$ along a vertical line, then $f(z)$ tends uniformly to L as z tends to x_0 through any cone of the form $\{\varepsilon < \arg(z - x_0) < \pi - \varepsilon\}$.



Stolz angle

5. Let $f(z)$ be a bounded analytic function on the open unit disk, and fix $e^{i\theta}$ on the unit circle. Show that if $f(re^{i\theta})$ has a limit L as r tends to 1, then $f(z)$ tends to L uniformly as z tends to $e^{i\theta}$ through any cone with vertex at $e^{i\theta}$, centered on the radius and with aperture strictly less than π . *Remark.* Such an approach sector is called a **Stolz angle** at $e^{i\theta}$.
6. Let D be a bounded domain, and let $f(z)$ be an analytic function from D into D . Denote by $f_n(z)$ the n th iterate of $f(z)$. Suppose that z_0 is an attracting fixed point for $f(z)$, that is, $f(z_0) = z_0$ and $|f'(z_0)| < 1$. Show that $f_n(z)$ converges uniformly on compact subsets of D to z_0 .
7. Let D be a bounded domain, and let $f(z)$ be an analytic function from D into D . Show that if $z_0 \in D$ is a fixed point for $f(z)$, then $|f'(z_0)| \leq 1$.
8. Let $\{f_n(z)\}$ be a sequence of analytic functions on a domain D . Suppose that $\iint_D |f_n(z)| dx dy \leq 1$ for $n \geq 1$.
- Show that $\{f_n(z)\}$ has a subsequence that converges normally to an analytic function $f(z)$ on D . *Hint.* To estimate $f(z)$, use the mean value property with respect to area (see Exercise III.4.1).
 - Show that $\iint_D |f(z)| dx dy \leq 1$.
 - Show that if $\iint_D |f_n(z) - f_m(z)| dx dy \rightarrow 0$ as $m, n \rightarrow \infty$, then $\iint_D |f_n(z) - f(z)| dx dy \rightarrow 0$ as $n \rightarrow \infty$.
9. Let D be a domain in the complex plane, and let $\mathcal{F}$ be the family of analytic functions $f(z)$ on D such that $|f(z)| \leq 1$ for $z \in D$. A

function $g(z)$ is an **extreme point** of $\mathcal{F}$ if it is not the midpoint of a line segment in $\mathcal{F}$, that is, if whenever $g = (g_0 + g_1)/2$ where $g_0, g_1 \in \mathcal{F}$, then $g_0 = g_1 = g$.

- (a) Show that the function $g \in B$ is an extreme point of $\mathcal{F}$ if and only if the only analytic function $h(z)$ satisfying $|g(z) \pm h(z)| \leq 1$ is $h = 0$.
- (b) Show that the function $g \in \mathcal{F}$ is an extreme point of $\mathcal{F}$ if and only if the only analytic function $h(z)$ satisfying $|g(z)| + |h(z)| \leq 1$ is $h = 0$. *Hint.* First show that if $|a \pm b| \leq 1$, then $|a| + |b|/2 \leq 1$.
- (c) Suppose that there is a nonconstant bounded analytic function on D , and fix $z_0 \in D$. Show that the Ahlfors function $G(z)$ associated with $z_0 \in D$ is an extreme point of $\mathcal{F}$.
- (d) Show that the Ahlfors function $G(z)$ associated with $z_0 \in D$ is unique, up to multiplication by a unimodular constant.
- (e) Let $L = \sum_{j,k} a_{jk} L_{jk}$ be a finite linear combination of linear functionals of the form $L_{jk}(f) = f^{(k)}(z_j)$, where the z_j 's are fixed points of D . Suppose there is a nonconstant bounded analytic function f on D such that $L(f) \neq 0$. Show that the extremal problem of maximizing $\operatorname{Re} L(f)$ over all $f \in \mathcal{F}$ has a unique extremal function $\varphi(z)$, which is an extreme point of $\mathcal{F}$.

6. Proof of the Riemann Mapping Theorem

Now we are ready for the proof of the Riemann mapping theorem. The key ingredients are Montel's theorem, Hurwitz's theorem, some elementary mapping properties of the square root function, and the fact that $\sqrt{z}$ is expanding on the open unit disk. We begin with the elementary mapping properties of the square root function.

Suppose that D is simply connected and that $D \neq \mathbb{C}$. Choose $a \in \mathbb{C} \setminus D$. By the characterization of simple connectivity (Section VIII.8), there is an analytic branch $g(z)$ of $\log(z - a)$ in D . Then $h(z) = e^{g(z)/2}$ is an analytic branch of $\sqrt{z - a}$ in D , and $h(z)^2 = z - a \neq 0$ in D . If $h(z_1) = h(z_2)$, then $z_1 = h(z_1)^2 + a = h(z_2)^2 + a = z_2$. Thus $h(z)$ is univalent, and $h(z)$ maps D conformally onto $h(D)$. Finally, note that if $w_0 \in h(D)$, then $-w_0 \notin h(D)$. Indeed, if $w_0 = h(z_0)$ and $-w_0 = h(z_1)$ for $z_0, z_1 \in D$, then $z_0 = h(z_0)^2 + a = w_0^2 + a = h(z_1)^2 + a = z_1$, which is impossible. We summarize.

Lemma. *Let D be a simply connected domain. Suppose $a \notin D$, and let $h(z)$ be an analytic branch of $\sqrt{z - a}$ in D . Then $h(z)$ is univalent on D , and further, $h(D)$ is disjoint from $-h(D)$.*

We remark that on the basis of the lemma we can show that any simply connected domain $D \neq \mathbb{C}$ is conformally equivalent to a subdomain of the unit disk, where we have hyperbolic geometry at our disposal. Indeed, if the disk $|w - w_0| \leq \varepsilon$ is contained in $h(D)$, then its negative is disjoint from $h(D)$, so that $|h(z) + w_0| > \varepsilon$ for $z \in D$, and $\varepsilon/(h(z) + w_0)$ maps D conformally onto a domain contained in the unit disk.

Our proof depends at least conceptually on hyperbolic geometry. Let $\rho(\zeta, \xi)$ be the hyperbolic metric in $\mathbb{D}$ (Section IX.3). The function ζ^2 , regarded as an analytic function from $\mathbb{D}$ to $\mathbb{D}$, is a contraction with respect to the hyperbolic metric, and it is a strict contraction on bounded subsets $\{|\zeta| \leq 1 - \varepsilon\}$ of the hyperbolic disk, in the sense that there is $c < 1$, depending on ε , such that $\rho(\zeta^2, \xi^2) \leq c\rho(\zeta, \xi)$ for $|\zeta|, |\xi| \leq 1 - \varepsilon$. Thus any branch of its inverse $\sqrt{\zeta}$ is expanding with respect to the hyperbolic metric, and it is strictly expanding on bounded subsets of the hyperbolic disk, in the sense that for any $r < 1$, there is $C > 1$ such that

$$\rho(\pm\sqrt{\zeta}, \pm\sqrt{\xi}) \geq C\rho(\zeta, \xi), \quad |\zeta|, |\xi| \leq r.$$

(For a precise value for C , see Exercise IX.3.9.) The expanding property of $\sqrt{\zeta}$ is the key to the following lemma.

Lemma. *Let D be a simply connected subdomain of $\mathbb{D}$ such that $0 \in D$. If $D \neq \mathbb{D}$, then there is a conformal map $\psi(\zeta)$ of D onto a subdomain of $\mathbb{D}$ such that $\psi(0) = 0$ and $|\psi'(0)| > 1$.*

Let $b \in \mathbb{D} \setminus D$, and let g be the conformal self-map of $\mathbb{D}$ that maps b to 0. Then $g(D)$ is a simply connected domain in the unit disk that does not contain 0. Hence there is an analytic branch $h(\xi)$ of $\sqrt{\xi}$ on $g(D)$, and further, as shown above, h is univalent on $g(D)$. Let f be a conformal self-map of $\mathbb{D}$ that maps $h(g(0))$ to 0, and set $\psi = f \circ h \circ g$. Then $\psi(\zeta)$ is a conformal map of D onto a subdomain of $\mathbb{D}$, and $\psi(0) = 0$. Since f and g are isometries in the hyperbolic metric, and h is a strict expansion near $g(0)$, ψ is a strict expansion near 0, that is, there is $C > 1$ such that $\rho(\psi(\zeta), 0) \geq C\rho(\zeta, 0)$ for $|\zeta| < \varepsilon$. Since $\rho(\psi(\zeta), 0) \sim 2|\psi(\zeta)|$ and $\rho(\zeta, 0) \sim 2|\zeta|$ as $\zeta \rightarrow 0$, we obtain

$$C \leq \frac{\rho(\psi(\zeta), 0)}{\rho(\zeta, 0)} \sim \frac{|\psi(\zeta)|}{|\zeta|} \rightarrow |\psi'(0)|,$$

and $|\psi'(0)| > 1$. This inequality can also be checked by expressing f and g explicitly and differentiating (Exercise 1).

Now we complete the proof of the Riemann mapping theorem by producing the Riemann map as an extremal function for an extremal problem similar to the problem considered in Section 2. We assume that D is a simply connected domain, $D \neq \mathbb{C}$. Fix $z_0 \in D$, and let $\mathcal{F}$ be the family of univalent(!) functions $f(z)$ on D such that $|f(z)| < 1$ for $z \in D$ and

$f(z_0) = 0$. The family $\mathcal{F}$ is nonempty, since if $h(z)$ maps D conformally onto a bounded domain, then the function $f(z) = \varepsilon(h(z) - h(z_0))$ is in $\mathcal{F}$ for $\varepsilon > 0$ small. We consider the extremal problem of maximizing $|f'(z_0)|$ over $f \in \mathcal{F}$. As before, set

$$A = \sup\{|f'(z_0)| : f \in \mathcal{F}\} > 0,$$

and let $\{f_n(z)\}$ be a sequence of functions in $\mathcal{F}$ such that $|f'_n(z_0)| \rightarrow A$. By Montel's theorem, the f_n 's have a subsequence that converges normally on D to an analytic function $\varphi(z)$. Clearly, $|\varphi(z)| \leq 1$, and $|\varphi'(z_0)| = A$. The functions f_n are univalent, so by Hurwitz's theorem (Section VIII.2), either φ is constant or φ is univalent. Since $\varphi'(z_0) \neq 0$, φ is nonconstant, and consequently, φ is univalent, mapping D onto a subdomain of $\mathbb{D}$. We claim that $\varphi(D) = \mathbb{D}$. Otherwise, we could apply the preceding Lemma to the domain $\varphi(D)$ and find a conformal map $\psi(\zeta)$ of $\varphi(D)$ onto a subdomain of $\mathbb{D}$ such that $\psi(0) = 0$ and $|\psi'(0)| > 1$. Then $\psi \circ \varphi \in \mathcal{F}$ would satisfy $|(\psi \circ \varphi)'(z_0)| = |\psi'(0)\varphi'(z_0)| > A$, contradicting the definition of A . This completes the proof.

We will give another proof of the Riemann mapping theorem in Section XV.5, based on the solvability of the Dirichlet problem.

Exercises for XI.6

1. Find explicitly the functions $f(\zeta)$ and $g(\zeta)$ used in the proof of the lemma. Show by computing the derivative that $|(f \circ h \circ g)'(0)| > 1$.
2. Let $\varphi(z)$ be the Riemann map of a simply connected domain D onto the open unit disk, normalized by $\varphi(z_0) = 0$ and $\varphi'(z_0) > 0$. Show that if $f(z)$ is any analytic function on D such that $|f(z)| \leq 1$ for $z \in D$, then $|f'(z_0)| \leq \varphi'(z_0)$, with equality only when $f(z)$ is a constant multiple of $\varphi(z)$. *Remark.* This shows that $\varphi(z)$ is the Ahlfors function of D corresponding to z_0 .
3. Let $\varphi(z)$ be the Riemann map of a simply connected domain D onto the open unit disk, normalized by $\varphi(z_0) = 0$ and $\varphi'(z_0) > 0$. Show that if $f(z)$ is any analytic function on D such that $|f(z)| \leq 1$ for $z \in D$, then $\operatorname{Re} f'(z_0) \leq \varphi'(z_0)$, with equality only when $f(z) = \varphi(z)$.
4. Let $\{D_m\}$ be an increasing sequence of simply connected domains, and let φ_m be the Riemann map of D_m onto the open unit disk $\mathbb{D}$, normalized so that $\varphi_m(z_0) = 0$ and $\varphi'_m(z_0) > 0$ for some fixed $z_0 \in D_1$. Let D be the union of the D_m 's. Show that if D is the entire complex plane, then the φ_m 's are eventually defined on any disk $\{|z| \leq R\}$ and converge there uniformly to 0. Otherwise, D is simply connected and the φ_m 's are eventually defined on each compact subset of D and converge there uniformly to the Riemann map φ of D onto $\mathbb{D}$ satisfying $\varphi(z_0) = 0$ and $\varphi'(z_0) > 0$.

5. Let $\{D_m\}$ be a decreasing sequence of simply connected domains, and suppose $w_0 \in D_m$ for all m . Let $g_m(z)$ be the conformal map of the open unit disk $\mathbb{D}$ onto D_m , normalized so that $g_m(0) = w_0$ and $g'_m(0) > 0$. Show that $g_m(z)$ converges normally on $\mathbb{D}$ to a function $g(z)$. If the distance from w_0 to the boundary of D_m tends to 0, then $g(z)$ is the constant function w_0 , and otherwise, $g(z)$ maps $\mathbb{D}$ conformally onto some simply connected domain D . Describe D in terms of the D_m 's.
6. Show that the function $\zeta \mapsto \zeta^2$ is not a strict contraction of the hyperbolic disk, that is, show that there is no constant $c < 1$ such that $\rho(\zeta^2, \xi^2) \leq c\rho(\zeta, \xi)$ for all $\zeta, \xi \in \mathbb{D}$. *Remark.* See Exercise IX.3.9.
7. Suppose $f : \mathbb{D} \rightarrow \mathbb{D}$ is an analytic function from the unit disk into itself with a fixed point at $z_0 \in \mathbb{D}$. Show that the stretching at z_0 of $f(z)$ in the hyperbolic metric is the same as the stretching at z_0 of $f(z)$ in the Euclidean metric,

$$\lim_{z \rightarrow z_0} \frac{\rho(f(z), z_0)}{\rho(z, z_0)} = \lim_{z \rightarrow z_0} \frac{|f(z) - z_0|}{|z - z_0|} = |f'(z_0)|.$$

XII

Compact Families of Meromorphic Functions

In Sections 1 and 2 we treat normal families of meromorphic functions. These are families that are sequentially compact when regarded as functions with values in the extended complex plane. We give two characterizations of normal families, Marty's theorem in Section 1 and the Zalcman lemma in Section 2. From the latter characterization we deduce Montel's theorem on compactness of families of meromorphic functions that omit three points, and we also prove the Picard theorems. Sections 3 and 4 constitute an introduction to iteration theory and Julia sets. In Section 3 we proceed far enough into the theory to see how Montel's theorem enters the picture and to indicate the fractal nature (self-similarity) of Julia sets. In Section 4 we relate the connectedness of Julia sets to the orbits of critical points. In Section 5 we introduce the Mandelbrot set, which has been called the "most fascinating and complicated subset of the complex plane."

1. Marty's Theorem

In this section we consider the convergence of sequences of meromorphic functions. Our aim is to give conditions guaranteeing compactness of families of meromorphic functions. Since functions may now assume the value ∞ , we must modify our notion of convergence. We do this by regarding meromorphic functions on a domain D as functions from D to the extended complex plane $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$, which we identify with a sphere via stereographic projection (Section I.3). When we identify the extended complex plane with the sphere, we refer to it also as the **Riemann sphere**. We will use distances on the sphere to measure distances between function values.

Let $\sigma(z, w)$ denote the distance from z to w in the spherical metric. Recall from Section IX.3 that $\sigma(z, w)$ is the length of the arc of the great circle on the Riemann sphere joining the points corresponding to z and w . The spherical metric is invariant under the transformations corresponding

to rotations of the sphere. Since the inversion $z \mapsto 1/z$ corresponds to a rotation of the sphere (by 180° around the x -axis), we have

$$(1.1) \quad \sigma(z, w) = \sigma(1/z, 1/w), \quad z, w \in \mathbb{C}.$$

On any fixed bounded subset of the complex plane, the spherical metric is equivalent to the Euclidean metric,

$$\frac{1}{B_R}|z - w| \leq \sigma(z, w) \leq B_R|z - w|, \quad |z|, |w| \leq R,$$

while on any fixed subset of the complex plane at a positive distance from the origin, the spherical distance from z to w is comparable to the Euclidean distance between the inverse points $1/z$ and $1/w$,

$$\frac{1}{C_\varepsilon} \left| \frac{1}{z} - \frac{1}{w} \right| \leq \sigma(z, w) \leq C_\varepsilon \left| \frac{1}{z} - \frac{1}{w} \right|, \quad |z|, |w| \geq \varepsilon.$$

We say that a sequence $\{f_n(z)\}$ of meromorphic functions on a domain D **converges normally** to $f(z)$ on D if the sequence converges uniformly on compact subsets of D to $f(z)$ in the spherical metric, that is, $\sigma(f_n(z), f(z))$ converges to zero uniformly on each compact subset of D as $n \rightarrow \infty$. On account of (1.1), we see that $\{f_n(z)\}$ converges normally to $f(z)$ if and only if $\{1/f_n(z)\}$ converges normally to $1/f(z)$. Here we declare $1/0 = \infty$ and $1/\infty = 0$, and we further allow the possibility of the constant function that is identically equal to ∞ .

Since the spherical and Euclidean metrics are equivalent on bounded subsets of the complex plane, the definition of normal convergence given above is consistent with our earlier definition of a normally convergent sequence of analytic functions. However, now we allow the possibility of the limit being ∞ .

Theorem. *If the sequence $\{f_n(z)\}$ of meromorphic functions on a domain D converges normally to $f(z)$, then either $f(z)$ is meromorphic on D or $f(z) \equiv \infty$. If a sequence $\{f_n(z)\}$ of analytic functions on D converges normally to $f(z)$, then either $f(z)$ is analytic on D or $f(z) \equiv \infty$.*

For the first statement, note that every point z for which $f(z) \neq \infty$ has a neighborhood on which the f_n 's are eventually uniformly bounded and converge uniformly in the Euclidean metric. Thus $f(z)$ is analytic on the set where $|f(z)| < \infty$. By the same token, since $1/f_n(z)$ converges normally to $1/f(z)$, also $1/f(z)$ is analytic wherever $f(z) \neq 0$. Either $1/f(z)$ is identically zero, in which case $f(z) \equiv \infty$; or else the zeros of $1/f(z)$ are isolated in D and they are poles of $f(z)$, in which case $f(z)$ is meromorphic on D .

For the second statement, suppose that the f_n 's are analytic with limit $f(z)$ satisfying $f(z_0) = \infty$ at some point. Then $1/f_n(z)$ is a meromorphic function with no zeros, and $1/f_n(z_0) \rightarrow 0$. Hurwitz's theorem (Sec-

tion VIII.3) shows that $1/f_n(z)$ converges uniformly to zero on some disk centered at z_0 , and consequently, $1/f(z) = 0$ on the disk. By the uniqueness principle, $1/f(z) = 0$ on D , and $f(z) = \infty$ on D .

We define a family $\mathcal{F}$ of meromorphic functions on D to be a **normal family** if every sequence in $\mathcal{F}$ has a subsequence that converges normally on D . In view of the preceding theorem, a family $\mathcal{F}$ of analytic functions on D is a normal family if and only if every sequence in $\mathcal{F}$ has either a subsequence that converges uniformly on compact subsets of D to an analytic function or a subsequence that converges uniformly on compact subsets of D to ∞ . The thesis version of Montel's theorem (Section XI.5) can be formulated in terms of normal families. It asserts that any family of analytic functions that is uniformly bounded on each compact set is a normal family.

The Arzelà-Ascoli theorem (Section XI.5) provides a characterization of normal families of meromorphic functions. A family $\mathcal{F}$ of meromorphic functions is a normal family if and only if the family $\mathcal{F}$ is equicontinuous, regarded as functions from D , with the Euclidean metric, to the extended complex plane $\mathbb{C}^*$, with the spherical metric. In order to exploit this condition, we study the spherical metric in more detail.

Recall (Section IX.3) that the spherical metric is $2|dz|/(1 + |z|^2)$. The spherical length of a curve $\gamma(t)$, $a \leq t \leq b$, is given by

$$\text{spherical length of } \gamma = \int_{\gamma} \frac{2|dz|}{1 + |z|^2} = \int_a^b \frac{2|\gamma'(t)| dt}{1 + |\gamma(t)|^2}.$$

If now $w = f(z)$ is a meromorphic function, then the spherical length of the image curve $f \circ \gamma$ is given by

$$\text{spherical length of } f \circ \gamma = \int_{\gamma} \frac{2|f'(z)|}{1 + |f(z)|^2} |dz|.$$

This leads us to define the **spherical derivative** of $f(z)$ to be

$$f^{\sharp}(z) = \frac{2|f'(z)|}{1 + |f(z)|^2}.$$

The spherical length of $f \circ \gamma$ is then simply

$$\text{spherical length of } f \circ \gamma = \int_{\gamma} f^{\sharp}(z) |dz|.$$

By taking γ to be a short straight line segment from z to $z + \Delta z$, we see that $f^{\sharp}(z)$ measures the expansion of $f(z)$, regarded as a map from the complex plane with the usual Euclidean metric to the extended complex plane with the spherical metric.

Since the spherical metric is invariant under the inversion $z \mapsto 1/z$, by (1.1), the spherical derivative is also invariant under the inversion,

$$(1.2) \quad (1/f)^{\sharp} = f^{\sharp}.$$

This identity can also be checked by direct calculation (Exercise 1). In dealing with the spherical metric on the range of $f(z)$, (1.2) allows us to replace $f(z)$ near its poles by $g(z) = 1/f(z)$, thereby reducing to the situation where $f(z)$ is analytic. This observation is the key to the following lemma and to Marty's theorem.

Lemma. *If $f_k(z) \rightarrow f(z)$ normally on D , then $f_k^\#(z) \rightarrow f^\#(z)$ uniformly on compact subsets of D .*

Indeed, if f is analytic at z_0 , then $f'_k \rightarrow f'$ uniformly in some neighborhood of z_0 , so $f_k^\# \rightarrow f^\#$ uniformly in some neighborhood of z_0 . If f is not analytic at z_0 , then $1/f$ is analytic at z_0 and $1/f_k \rightarrow 1/f$ normally. Again $f_k^\# = (1/f_k)^\#$ converges uniformly to $f^\# = (1/f)^\#$ in some neighborhood of z_0 .

Theorem (Marty's Theorem). *A family $\mathcal{F}$ of meromorphic functions is normal on a domain D if and only if the spherical derivatives $\{f^\#(z) : f \in \mathcal{F}\}$ are uniformly bounded on each compact subset of D .*

Suppose the spherical derivatives are uniformly bounded near $z_0 \in D$, say $f^\#(z) \leq C$ for $|z - z_0| < r$ and $f \in \mathcal{F}$. If $|z_1 - z_0| < r$, and γ is the straight line segment from z_0 to z_1 , then the spherical distance from $f(z_0)$ to $f(z_1)$ is estimated by

$$\sigma(f(z_0), f(z_1)) \leq \int_{\gamma} f^\#(z) |dz| \leq C|z_0 - z_1|.$$

Since this estimate is independent of the function $f \in \mathcal{F}$, regarded as functions from the Euclidean to the spherical metric the family $\mathcal{F}$ is equicontinuous at z_0 . The Arzelà-Ascoli theorem then implies that $\mathcal{F}$ is normal. On the other hand, if the spherical derivatives of the functions in $\mathcal{F}$ are not uniformly bounded on compact sets, there are $f_k \in \mathcal{F}$ such that the maximum of $f_k^\#$ over some compact set tends to $+\infty$. By the lemma, $\{f_k\}$ cannot have a normally convergent subsequence, so $\mathcal{F}$ is not normal.

The definition of a normal family of meromorphic functions on a domain D can be extended to include domains D that contain ∞ . We say that a family $\mathcal{F}$ of meromorphic functions on a domain D in the extended complex plane is a **normal family of meromorphic functions** on D if $\mathcal{F}$ is a normal family on $D \setminus \{\infty\}$, and if further the functions $g(w) = f(1/w)$, $1/w \in D$, form a normal family of meromorphic functions in some disk containing $w = 0$. Thus the notion of a normal family is "local." Marty's theorem remains valid for domains containing ∞ .

Exercises for XII.1

1. Verify the identity $(1/f)^\sharp(z) = f^\sharp(z)$ by direct calculation.
2. Show that $(g \circ f)^\sharp(z) = g^\sharp(f(z))|f'(z)|$. Interpret the identity in terms of stretching with respect to Euclidean and spherical metrics.
3. Show that the functions $z/(z+\varepsilon)$, $0 < \varepsilon \leq 1$, form a normal family of meromorphic functions on $\mathbb{C}^* \setminus \{0\}$. Show that the functions $g_\varepsilon(w) = f_\varepsilon(1/w)$ form a normal family of meromorphic functions on $\mathbb{C}$. Find $\lim f_\varepsilon^\sharp(z)$ and $\lim g_\varepsilon^\sharp(w)$ as $\varepsilon \rightarrow 0$.
4. Show that the functions $1/(z+\varepsilon)$, $0 < \varepsilon \leq 1$, form a normal family of meromorphic functions on $\mathbb{C}^* \setminus \{0\}$, but they do *not* form a normal family of meromorphic functions on $\mathbb{C}$.
5. Fix $M > 1$, and let $\{f_n(z)\}$ be a sequence of meromorphic functions on a domain D . Show that $f_n(z)$ converges normally to $f(z)$ if and only if $f_n(z)$ converges uniformly to $f(z)$ on compact subsets of the open set $\{|f(z)| < M\}$, and $1/f_n(z)$ converges uniformly to $1/f(z)$ on compact subsets of the open set $\{|f(z)| > 1/M\}$.
6. Let E be a compact connected subset of the complex plane that contains more than one point. Show that the family of meromorphic functions on a domain D that omits E (that is, with range in $\mathbb{C}^* \setminus E$) is a normal family of meromorphic functions.
7. Let $g(z)$ be a nonconstant analytic function on a domain U , let $V = g(U)$, and let $\mathcal{F}$ be a family of meromorphic functions on V . Show that the family $\mathcal{F}$ is normal on V if and only if the family of compositions $\{f \circ g : f \in \mathcal{F}\}$ is normal on U .
8. Let $\{f_n(z)\}$ be a sequence of rational functions that converges normally to $f(z)$ on the extended complex plane $\mathbb{C}^*$. Show that $f_n(z)$ has the same degree as $f(z)$ for n large. (See Exercise VIII.4.2 for the definition of the degree of a rational function.)
9. Let $\mathcal{F}$ be a family of meromorphic functions on a domain D . Suppose there is an increasing function $\psi(t)$, $t \geq 0$, such that $|f'(z)| \leq \psi(|f(z)|)$ for all $z \in D$ and $f \in \mathcal{F}$. Show that $\mathcal{F}$ is normal on D . *Remark.* This is **Royden's theorem**. For the proof, assume that $\psi(t)$ is smooth and increasing, and $\psi(t) \geq 1 + t^2$. Consider the metric in which the distance from z to w is the infimum of $\int 1/\psi(|\zeta|) |d\zeta|$ over all smooth paths from z to w . Show that this metric is equivalent to the spherical metric.
10. Show that the family of analytic functions on a domain satisfying $|f'(z)| \leq e^{|f(z)|}$ is normal.

2. Theorems of Montel and Picard

As an application of Marty's theorem, we give a proof of Montel's theorem characterizing normal families of meromorphic functions. Another proof of Montel's theorem will be outlined at the end of Chapter XVI, based on a modular function that arises as a covering map. We begin with the following lemma. Since the converse is trivial, it actually provides a characterization of normal families.

Theorem (Zalcman's Lemma). *Suppose $\mathcal{F}$ is a family of meromorphic functions on a domain D that is not normal. Then there are points $z_n \in D$ converging to a point of D , numbers $\rho_n > 0$ converging to 0, and functions $f_n \in \mathcal{F}$ such that the scaled functions $g_n(\zeta) = f_n(z_n + \rho_n \zeta)$ converge normally to a nonconstant meromorphic function $g(\zeta)$ on $\mathbb{C}$ satisfying $g^\sharp(0) = 1$ and $g^\sharp(\zeta) \leq 1$ for $\zeta \in \mathbb{C}$.*

The proof is tricky. By Marty's theorem, there are sequences $\{w_n\}$ in a compact subset of D and $f_n \in \mathcal{F}$ such that $f_n^\sharp(w_n) \rightarrow +\infty$. We can assume that $w_n \rightarrow 0 \in D$, and we assume for convenience that the closed unit disk $\{|z| \leq 1\}$ is contained in D . Define

$$R_n = \max_{|z| \leq 1} f_n^\sharp(z)(1 - |z|).$$

Since $w_n \rightarrow 0$ and $f_n^\sharp(w_n) \rightarrow \infty$, we have $R_n \rightarrow \infty$. Suppose $f_n^\sharp(z)(1 - |z|)$ attains its maximum at the point z_n ,

$$R_n = f_n^\sharp(z_n)(1 - |z_n|).$$

Since $f_n^\sharp(z_n) \geq R_n$, also $f_n^\sharp(z_n) \rightarrow \infty$. Define

$$\rho_n = 1/f_n^\sharp(z_n);$$

then $\rho_n \rightarrow 0$. The disk centered at z_n of radius $1 - |z_n| = \rho_n R_n$ is contained in the unit disk and hence in D . We parametrize it with a parameter ζ , by

$$\zeta \mapsto z_n + \rho_n \zeta, \quad |\zeta| < R_n.$$

Define the scaled function $g_n(\zeta)$ by

$$g_n(\zeta) = f_n(z_n + \rho_n \zeta), \quad |\zeta| < R_n.$$

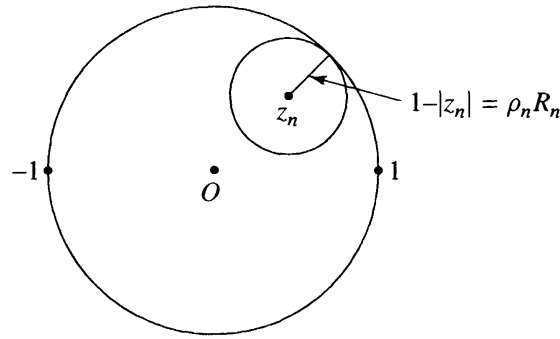
Since $R_n \rightarrow \infty$, these functions are eventually defined on any compact set. From the chain rule $(f \circ h)^\sharp(\zeta) = f^\sharp(h(\zeta))|h'(\zeta)|$ (Exercise 1.2) with $h(\zeta) = z_n + \rho_n \zeta$, we obtain

$$g_n^\sharp(\zeta) = \rho_n f_n^\sharp(z_n + \rho_n \zeta), \quad |\zeta| < R_n.$$

Now fix $R > 0$. If n is so large that $R_n > R$, then $g_n(\zeta)$ is defined on the disk $\{|\zeta| < R\}$, and since $f_n^\#(z_n + \rho_n \zeta)(1 - |z_n + \rho_n \zeta|) \leq R_n$ we have

$$\begin{aligned} g_n^\#(\zeta) &\leq \rho_n \frac{R_n}{1 - |z_n + \rho_n \zeta|} \leq \frac{\rho_n R_n}{1 - |z_n| - \rho_n R} \\ &= \frac{\rho_n R_n}{\rho_n R_n - \rho_n R} = \frac{1}{1 - R/R_n}, \quad |\zeta| < R. \end{aligned}$$

By Marty's theorem, the g_n 's for n large form a normal family on the disk $\{|\zeta| < R\}$. Passing to a subsequence, we can then assume that $\{g_n\}$ converges normally on $\mathbb{C}$ to a meromorphic function $g(\zeta)$. Since for each fixed R , $1/(1 - R/R_n) \rightarrow 1$, the estimate on $g_n^\#(\zeta)$ shows that $g^\#(\zeta) \leq 1$ for all $\zeta \in \mathbb{C}$. Since $g_n^\#(0) = \rho_n f_n^\#(z_n) = 1$, $g^\#(0) = 1$.



Suppose that $f(z)$ is meromorphic for $0 < |z - z_0| < r$. A value $w_0 \in \mathbb{C}^*$ is an **omitted value** of $f(z)$ at z_0 if there is $\delta > 0$ such that $f(z) \neq w_0$ for $0 < |z - z_0| < \delta$. Thus w_0 is *not* an omitted value of $f(z)$ at z_0 if and only if there is a sequence, $z_n \rightarrow z_0$, $z_n \neq z_0$, such that $f(z_n) = w_0$. An omitted value of $f(z)$ at ∞ is defined similarly.

Example. The function $e^{1/z}$ omits the two values 0 and ∞ at $z_0 = 0$. The function e^z omits the two values 0 and ∞ at ∞ .

Theorem (Montel's Theorem). *A family $\mathcal{F}$ of meromorphic functions on a domain D that omits three values is normal.*

Since normality is a local property, we can assume that the domain D is the open unit disk $\{|z| < 1\}$. By composing the functions in $\mathcal{F}$ with a fractional linear transformation, we can assume that the omitted values are 0, 1, and ∞ . Since the functions in $\mathcal{F}$ are then analytic and nonzero on the unit disk, they have analytic roots of all orders. Let $\mathcal{F}_k$ be the family of all 2^k th roots of functions in $\mathcal{F}$. Evidently $\mathcal{F}_k$ is normal if and only if $\mathcal{F}$ is normal. The functions in $\mathcal{F}_k$ omit the values 0, ∞ , and all 2^k th roots of unity.

We will argue by contradiction. Suppose $\mathcal{F}$ is *not* normal. Then $\mathcal{F}_k$ is not normal. Let $G_k(\zeta)$ be the entire function from the Zalcman lemma, so that $G_k^\#(\zeta) \leq 1$, $G_k^\#(0) = 1$, and G_k is a limit of restrictions of functions in $\mathcal{F}_k$,

appropriately scaled. Since the functions in $\mathcal{F}_k$ omit the 2^k th roots of unity, so do their scaled restrictions, and from Hurwitz's theorem (Section VIII.3), so does any nonconstant normal limit. Thus G_k omits the 2^k th roots of unity. Now, Marty's theorem shows that $\{G_k\}$ is a normal family. Let G be any normal limit of a subsequence of the G_k 's. Then $G^\sharp(0) = 1$, so that G is nonconstant, and by Hurwitz's theorem again, G omits all 2^k th roots of unity, this for all $k \geq 1$. Since G is an open mapping, G omits the unit circle. Thus either $|G| < 1$ or $|G| > 1$. Applying Liouville's theorem to G in the first case and to $1/G$ in the second, we conclude that G is constant. This contradiction establishes the theorem.

The following theorem is a substantial generalization of the Casorati-Weierstrass theorem.

Theorem (Picard's Big Theorem). *Suppose $f(z)$ is meromorphic on a punctured neighborhood $\{0 < |z - z_0| < \delta\}$ of z_0 . If $f(z)$ omits three values at z_0 , then $f(z)$ extends to be meromorphic at z_0 .*

For the proof, we may assume that $z_0 = 0$ and that $f(z)$ omits the values 0 and ∞ on the punctured disk. Then $f(z)$ is analytic on the punctured disk. Let $\{\varepsilon_n\}$ be a sequence that decreases to 0, and define

$$g_n(z) = f(\varepsilon_n z), \quad 0 < |z| < \delta.$$

Then $\{g_n\}$ omits three values, including 0 and ∞ . By Montel's theorem, $\{g_n\}$ is a normal family. Passing to a subsequence, we can assume that $g_n(z)$ converges normally to $g(z)$ for $0 < |z| < \delta$. Assume first that $g(z)$ is not identically ∞ . Then $g(z)$ is analytic for $0 < |z| < \delta$. Fix ρ , $0 < \rho < \delta$, and choose M such that $|g(z)| < M$ for $|z| = \rho$. Then for large n we have $|g_n(z)| < M$ for $|z| = \rho$, and consequently, $|f(z)| < M$ for $|z| = \varepsilon_n \rho$. By the maximum principle, $|f(z)| < M$ for $\varepsilon_n \rho \leq |z| \leq \rho$, this for all large n . These annuli increase to a punctured neighborhood of 0 on which $|f(z)| < M$. By Riemann's theorem on removable singularities, $f(z)$ extends to be analytic at 0, and the theorem is proved in this case. If, on the other hand, $g(z) \equiv \infty$, then we apply the above argument to $1/f(z)$. The argument shows that $1/f(z)$ extends analytically to 0, so in this case $f(z)$ extends meromorphically to have a pole at 0.

Theorem (Picard's Little Theorem). *A nonconstant entire function assumes every value in the complex plane with at most one exception.*

In other words, an entire function that omits two (finite) values is constant. To see this, we regard ∞ as an isolated singularity of the function. The function omits two finite values and the value ∞ at the isolated singularity at ∞ . By Picard's big theorem, ∞ is a removable singularity or pole. Thus the function is a polynomial, and since any nonconstant polynomial assumes all finite values, the function is constant.

Example. The entire function e^z omits the value 0. The entire function $\sin z$ does not omit any (finite) value.

Example. The entire function ze^z does not omit any (finite) value. However, it does omit the value 0 at ∞ .

Exercises for XII.2

1. Show that $e^{1/z} + e^{-1/z}$ omits only the value ∞ at $z = 0$.
2. Show that the meromorphic function $e^{1/z} + 1/(1 - e^{1/z})$ on $\mathbb{C} \setminus \{0\}$ does not omit any value at $z = 0$.
3. Show that for $\alpha > 0$ and $0 < \beta < 1$, there are constants $C(\alpha, \beta) > 0$ with the following property. If $\mathcal{F}$ is a family of analytic functions on the open unit disk that omits the values 0 and 1, and if $|f(0)| < \alpha$ for all $f \in \mathcal{F}$, then $|f(z)| \leq C(\alpha, \beta)$ for all $f \in \mathcal{F}$ and $|z| \leq \beta$.
Remark. This is **Schottky's theorem**.
4. Let $f(z)$ be analytic on the punctured disk $\{0 < |z| < 1\}$, and define $f_n(z) = f(z/n)$, $n \geq 1$. Show that $\{f_n(z)\}$ is a normal family on the punctured disk if and only if the singularity of $f(z)$ at 0 is removable or a pole.
5. Let E_0, E_1, E_2 be three disjoint compact subsets of the Riemann sphere, and let $\mathcal{F}$ be a family of meromorphic functions on a domain D such that each $f \in \mathcal{F}$ omits at least one point of each of the three sets. Show that $\mathcal{F}$ is a normal family.
6. Let $\mathcal{G}$ be the family of univalent analytic functions on a fixed domain D . (a) Show that $\mathcal{G}$ is not normal. (b) Show that the family of functions in $\mathcal{G}$ that omit 0 is normal. (c) Show that the family of derivatives of functions in $\mathcal{G}$ is normal. (d) Show that for fixed $z_0 \in D$ and $M > 0$ the family of functions $f \in \mathcal{G}$ satisfying $|f'(z_0)| \leq M$ is normal.
7. Let $\mathcal{S}$ denote the family of univalent functions $f(z)$ on the open unit disk $\mathbb{D}$, normalized by $f(0) = 0$ and $|f'(0)| = 1$. (a) Show that there exists $\kappa > 0$ such that the image of the open unit disk $\mathbb{D}$ under any $f \in \mathcal{S}$ includes the open disk centered at 0 of radius κ . (b) Show that the function $f(z) = z/(1 - z)^2$ belongs to $\mathcal{S}$ and maps $\mathbb{D}$ onto the complex plane slit along the negative real axis from $-\frac{1}{4}$ to $-\infty$. Conclude that $\kappa \leq \frac{1}{4}$. *Remark.* The theorem in (a) was first proved by P. Koebe. L. Bieberbach showed that the estimate holds with $\kappa = \frac{1}{4}$, and this estimate is sharp. The theorem is known as **Koebe's one-quarter theorem**, and the function in (b) is referred to as the **Bieberbach function**.

8. Show that there is a constant $\beta > 0$ with the following property. If $f(z)$ is an analytic function on the open unit disk $\mathbb{D}$ such that $f(0) = 0$ and $f'(0) = 1$, there is a subdisk $D \subset \mathbb{D}$ such that $f(z)$ is one-to-one on D and $f(D)$ contains a disk of radius β . *Remark.* This is **Bloch's theorem**, and the optimal (largest) constant β is **Bloch's constant**.
9. Give a proof of Royden's theorem (Exercise 1.9) based on the Zalcman lemma.
10. A family $\mathcal{F}$ of meromorphic functions on a domain D is **normal at** $z_0 \in D$ if $\mathcal{F}$ is normal on some open disk centered at z_0 . Show that $\mathcal{F}$ is normal at z_0 if and only if whenever $\{z_n\}$ is a sequence in D that converges to z_0 and $\rho_n \rightarrow 0$, then every sequence in $\mathcal{F}$ has a subsequence $\{f_n(z)\}$ for which the corresponding scaled functions $g_n(\zeta) = f_n(z_n + \rho_n \zeta)$ converge normally to a constant (possibly ∞).
11. Let $\mathcal{F}$ be a family of meromorphic functions on a domain D that is not normal at $z_0 \in D$, and suppose that $f_n \in \mathcal{F}$, $z_n \rightarrow z_0$, and $\rho_n \rightarrow 0$ are such that the scaled functions $g_n(\zeta) = f_n(z_n + \rho_n \zeta)$ converge normally to a nonconstant meromorphic function $g(\zeta)$. Let $\zeta_0 \in \mathbb{C}$, $w_0 = g(\zeta_0)$. (a) Show that there is a sequence $\xi_n \rightarrow z_0$ such that $f_n(\xi_n) = w_0$. (b) Show that if ζ_0 is not a critical point of $g(\zeta)$, then $f_n^\#(\xi_n) \rightarrow \infty$. (c) Show that if $\psi(z)$ is a meromorphic function defined near w_0 such that $\psi(w_0) = z_0$, then there is a sequence $\eta_n \rightarrow z_0$ such that η_n is a fixed point of $\psi \circ f_n$, that is, $\psi(f_n(\eta_n)) = \eta_n$. (d) Show that if ζ_0 is not a critical point of $g(\zeta)$, and if w_0 is not a critical point of $\psi(w)$, then $(\psi \circ f_n)^\#(\eta_n) \rightarrow \infty$.

3. Julia Sets

One of the early applications of Montel's theorem was to complex dynamics, the study of the behavior of the iterates of an analytic or meromorphic function. Julia and Fatou used Montel's theorem as a key tool for studying the iterates of a rational function. One of their main ideas was to understand the dynamical behavior of the iterates by splitting the extended complex plane $\mathbb{C}^*$ into two invariant subsets, on one of which (the Fatou set) the iterates are well behaved, and on the other of which (the Julia set) their behavior is chaotic. We will derive some basic facts about Fatou and Julia sets in this section and the next.

Let U be a domain in the extended complex plane $\mathbb{C}^*$, and let $f(z)$ be an analytic map from U to U . In other words, $f(z)$ is an analytic function on U (meromorphic if $\infty \in U$) whose range is contained in U . For the remainder of this chapter it will be convenient to denote the n th iterate

$f(f(\cdots(f(z))\cdots))$ (n times) of $f(z)$ by $f^n(z)$. *Danger!* This should not be confused with the n th power $f(z)^n$ of $f(z)$.

Example. If $f(z) = z + 1$, then $f^n(z) = z + n$.

Example. If $f(z) = \lambda z$, then $f^n(z) = \lambda^n z$.

Example. If $f(z) = z^d$, then $f^n(z) = z^{d^n}$.

Example. If $f(z)$ is a polynomial of degree d , then $f^n(z)$ is a polynomial of degree d^n .

In general it is not possible to express the iterates of $f(z)$ in a simple form. An important method for obtaining effective information on the behavior of iterates is to make a “change of variable” in order to express the function in a more tractable form, such as one of the forms above, with respect to the new variable. Such a change of variable is called an **analytic conjugation**. It is defined formally as follows.

Let $f(z)$ be an analytic map from U into U , and $g(\zeta)$ an analytic map of V into V . We say that $f(z)$ and $g(\zeta)$ are **conjugate** if there is a conformal map $\zeta = \varphi(z)$ of U onto V such that

$$\varphi(f(z)) = g(\varphi(z)), \quad z \in U.$$

Thus $\varphi \circ f = g \circ \varphi$. We express this by saying that the following diagram commutes:

$$\begin{array}{ccc} U & \xrightarrow{f} & U \\ \varphi \downarrow & & \varphi \downarrow \\ V & \xrightarrow{g} & V. \end{array}$$

We can regard $f(z)$ and $g(\zeta)$ as the same analytic function, expressed in different coordinate systems. If φ conjugates $f(z)$ to $g(\zeta)$, then φ also conjugates each iterate of $f(z)$ to the corresponding iterate of $g(\zeta)$,

$$\varphi(f^n(z)) = g^n(\varphi(z)), \quad z \in U, n \geq 1.$$

Further, the inverse of $\varphi(z)$ conjugates $g(\zeta)$ to $f(z)$, $\varphi^{-1} \circ g = f \circ \varphi^{-1}$.

Example. Every polynomial $P(z) = Az^d + \cdots$ of degree $d \geq 2$ is conjugate to a monic polynomial. Indeed, if we take $\zeta = \varphi(z) = cz$ and $Q = \varphi \circ P \circ \varphi^{-1}$, we obtain

$$\begin{aligned} Q(\zeta) &= \varphi(P(z)) = cP(z) = cAz^d + \cdots \\ &= c^{1-d}A\zeta^d + \text{lower order.} \end{aligned}$$

We take c such that $c^{1-d}A = 1$, and we have conjugated $P(z)$ to a monic polynomial $Q(\zeta)$ of degree d .

Example. Consider $P(z) = z^2 - 2$. Let $z = \psi(\zeta) = \zeta + 1/\zeta$ be the conformal map from $\{|\zeta| > 1\}$ to the slit z -plane $\mathbb{C} \setminus [-2, 2]$. We compute that

$$P(\psi(\zeta)) = \left(\zeta + \frac{1}{\zeta}\right)^2 - 2 = \zeta^2 + \frac{1}{\zeta^2} = \psi(\zeta^2).$$

Thus $\varphi = \psi^{-1}$ conjugates $P(z)$ on $\mathbb{C} \setminus [-2, 2]$ to the map $\zeta \mapsto \zeta^2$ on the exterior of the closed unit disk.

From the conjugation identity $\varphi(f(z)) = g(\varphi(z))$ we see that $f(z_0) = z_0$ if and only if $g(\varphi(z_0)) = \varphi(z_0)$, that is, z_0 is a fixed point of $f(z)$ if and only if $\varphi(z_0)$ is a fixed point of $g(\zeta)$. Differentiation of the conjugation identity yields $\varphi'(f(z))f'(z) = g'(\varphi(z))\varphi'(z)$, from which it follows that $f'(z_0) = 0$ if and only if $g'(\varphi(z_0)) = 0$. We say that fixed points and critical points are “conjugation invariants.”

Example. Any fractional linear transformation $f(z)$ with exactly two fixed points can be conjugated to the multiplication $g(\zeta) = \lambda\zeta$ for some complex number $\lambda \neq 0$. Indeed, let $\varphi(z)$ be a fractional linear transformation that maps the fixed points z_0 and z_1 of $f(z)$ to 0 and ∞ , respectively. Then $\zeta = \varphi(z)$ conjugates f to the fractional linear transformation $g = \varphi \circ f \circ \varphi^{-1}$. Since $g(\zeta)$ now has fixed points at 0 and ∞ , $g(\zeta)$ must have the form $\lambda\zeta$. A similar argument shows that if $f(z)$ is a fractional linear transformation with only one fixed point, then $f(z)$ can be conjugated to the translation $g(\zeta) = \zeta + 1$. In this case we take $\varphi(z)$ to be the fractional linear transformation that maps the fixed point of $f(z)$ to ∞ , some other specified point z_0 to 0, and $f(z_0)$ to 1. (See Exercise II.7.12.)

Now let $f(z)$ be a rational function, regarded as an analytic map from $\mathbb{C}^*$ to $\mathbb{C}^*$. The **Fatou set** of $f(z)$, denoted by $\mathcal{F} = \mathcal{F}(f)$, consists of all points $z_0 \in \mathbb{C}^*$ that have an open neighborhood W such that the restrictions of the iterates of $f(z)$ to W form a normal family of analytic functions on W . In this case, the Fatou set of f contains along with z all points in the open set W , so the Fatou set is open. The **Julia set** of $f(z)$, denoted by $\mathcal{J} = \mathcal{J}(f)$, is the complement of the Fatou set, $\mathcal{J} = \mathbb{C}^* \setminus \mathcal{F}$. Thus $\mathcal{J}$ is a closed subset of $\mathbb{C}^*$.

Example. For $P(z) = z^2$ we have $P^n(z) = z^{2^n}$. Thus the iterates $\{P^n(z)\}$ converge normally to ∞ on $\{|z| > 1\}$ and they converge normally to 0 on $\{|z| < 1\}$. If W is any disk containing a point z_0 with $|z_0| = 1$, then the iterates $P^n(z)$ converge to both 0 and ∞ on nonempty open subsets of W , so the iterates do not form a normal family on W . Thus the Julia set of $P(z)$ coincides with the unit circle, and the Fatou set of $P(z)$ is the complement of the unit circle.

Theorem. *The Fatou set and the Julia set of a rational function $f(z)$ are invariant, that is, $f(\mathcal{F}) \subseteq \mathcal{F}$ and $f(\mathcal{J}) \subseteq \mathcal{J}$.*

Since $\mathcal{F}$ and $\mathcal{J}$ are complementary sets, the theorem amounts to the assertion that $f(\mathcal{F}) \subseteq \mathcal{F}$ and $f^{-1}(\mathcal{F}) \subseteq \mathcal{F}$, that is, that $\mathcal{F}$ is **completely invariant**. Thus we must show that $z_0 \in \mathcal{F}$ if and only if $f(z_0) \in \mathcal{F}$. One direction is trivial. That $f(z_0) \in \mathcal{F}$ implies $z_0 \in \mathcal{F}$ follows from the observation that if a subsequence of iterates $\{f^{n_k-1}(z)\}$ converges normally on a disk V containing $f(z_0)$, then the compositions $\{f^{n_k}(z)\}$ converge normally on the open set $f^{-1}(V)$ containing z_0 . The less obvious direction, that $z_0 \in \mathcal{F}$ implies $f(z_0) \in \mathcal{F}$, follows from the observation that if a subsequence of iterates $\{f^{n_k+1}(z)\}$ converges normally on a disk U containing z_0 , then $\{f^{n_k}(z)\}$ converges normally on $f(U)$, which is an open set containing $f(z_0)$. (See Exercise 1.7.)

For simplicity we focus now on a monic polynomial $P(z)$ of degree $d \geq 2$. The main feature of $P(z)$ that we will exploit is its attracting fixed point at $z = \infty$. Since $P(z) \sim z^d$ as $z \rightarrow \infty$, there is $R > 0$ such that $|P(z)| > 2|z|$ for $|z| \geq R$. The iterates $P^n(z)$ then converge uniformly to ∞ on the exterior domain $U_R = \{|z| > R\}$. In particular, $U_R \subset \mathcal{F}$.

The **basin of attraction** of ∞ , denoted by $A(\infty)$, is defined to be the set of all z such that $P^n(z) \rightarrow \infty$. This occurs if and only if $P^n(z) \in U_R$ for large n . We denote by $P^{-n}(U_R)$ the set of z such that $P^n(z) \in U_R$. This is the inverse image of U_R under $P^n(z)$, so it is open. Since $A(\infty)$ is the union of the open sets $P^{-n}(U_R)$, $A(\infty)$ is also open. Since $P^k(z)$ tends to ∞ as $k \rightarrow \infty$ uniformly on $P^{-n}(U_R)$, each $P^{-n}(U_R)$ is contained in $\mathcal{F}$, and $A(\infty) \subseteq \mathcal{F}$.

Theorem. *Let $P(z)$ be a polynomial of degree $d \geq 2$. The basin of attraction $A(\infty)$ of ∞ is an open connected subset of $\mathbb{C}^*$ containing ∞ . The Julia set $\mathcal{J}$ of $P(z)$ coincides with the boundary of $A(\infty)$, and it is a nonempty compact subset of $\mathbb{C}$. Each bounded component of $\mathbb{C} \setminus \mathcal{J}$ is simply connected.*

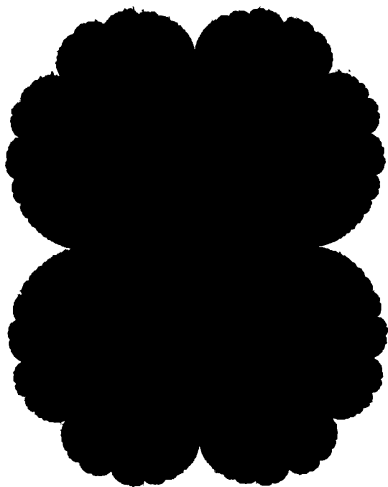
The proof is straightforward. First note that $A(\infty)$ is completely invariant, that is, $P(A(\infty)) \subseteq A(\infty)$ and $P^{-1}(A(\infty)) \subseteq A(\infty)$. Thus $P(\partial A(\infty))$ is disjoint from $A(\infty)$, and $P(\partial A(\infty)) \subseteq \partial A(\infty)$. Then also $P^n(\partial A(\infty)) \subseteq \partial A(\infty)$ for all $n \geq 1$. In particular, the iterates $P^n(z)$ are uniformly bounded on $\partial A(\infty)$. By the maximum principle, the iterates $P^n(z)$ are uniformly bounded on the bounded components of the complement $\mathbb{C} \setminus \partial A(\infty)$ of $\partial A(\infty)$. Thus no bounded component of $\mathbb{C} \setminus \partial A(\infty)$ is iterated to ∞ , and $A(\infty)$ consists of just one connected component, the unbounded component of $\mathbb{C} \setminus \partial A(\infty)$. By Montel's theorem, the iterates $P^n(z)$ form a normal family on each bounded component of the open set $\mathbb{C} \setminus \partial A(\infty)$, so that all these components belong to the Fatou set $\mathcal{F}$, and the Julia set $\mathcal{J}$ is contained in $\partial A(\infty)$. If $z_0 \in \partial A(\infty)$, and U is any open disk containing z_0 , then $\{P^n(z)\}$ converges normally to ∞ on $U \cap A(\infty)$, while the sequence is bounded at z_0 . Thus no subsequence of $\{P^n(z)\}$ can converge normally on U , and so $z_0 \in \mathcal{J}$. Thus $\mathcal{J} = \partial A(\infty)$. Since the

polynomial $P(z)$ has degree $d \geq 2$, it has at least one fixed point $z_0 \in \mathbb{C}$. Evidently, $z_0 \notin A(\infty)$, so $A(\infty) \neq \mathbb{C}^*$, and $\mathcal{J} = \partial A(\infty)$ is not empty. Finally, since $A(\infty)$ is connected, also $A(\infty) \cup \partial A(\infty)$ is connected, and each bounded component of the Fatou set is simply connected. (See Section VIII.8.) This completes the proof.

Example. For $P(z) = z^2$, the basin of attraction of ∞ is the exterior of the unit circle, $A(\infty) = \{|z| > 1\}$. The other component of the Fatou set is the open unit disk, which is the basin of attraction of the fixed point 0.

Example. Let $P(z) = z^2 - 2$. We have seen that $P^n(z) \rightarrow \infty$ for all $z \in \mathbb{C} \setminus [-2, 2]$. On the other hand, if $-2 \leq x \leq 2$, then $-2 \leq P(x) \leq 2$, so the interval $[-2, 2]$ is invariant under $P(z)$ and disjoint from $A(\infty)$. Thus $A(\infty) = \mathbb{C}^* \setminus [-2, 2]$, and the Julia set is $\mathcal{J} = [-2, 2]$. In this case, the Fatou set coincides with $A(\infty)$.

These examples of “smooth” Julia sets are the exception rather than the rule. It turns out that the only values of c for which the Julia set of $z^2 + c$ lies on a smooth curve are the values $c = 0$ and $c = -2$ treated above. Several filled-in Julia sets for other values of the parameter c are depicted below. It can be shown that the Julia set of $z^2 - 0.6$ (next page) is a simple closed Jordan curve that is nowhere differentiable. This is true of all values of c in the “principal cardioid of the Mandelbrot set,” except $c = 0$. We will show in the next section that the Julia set of $z^2 + 0.251$ (see below) is totally disconnected, that is, it contains no continuum.

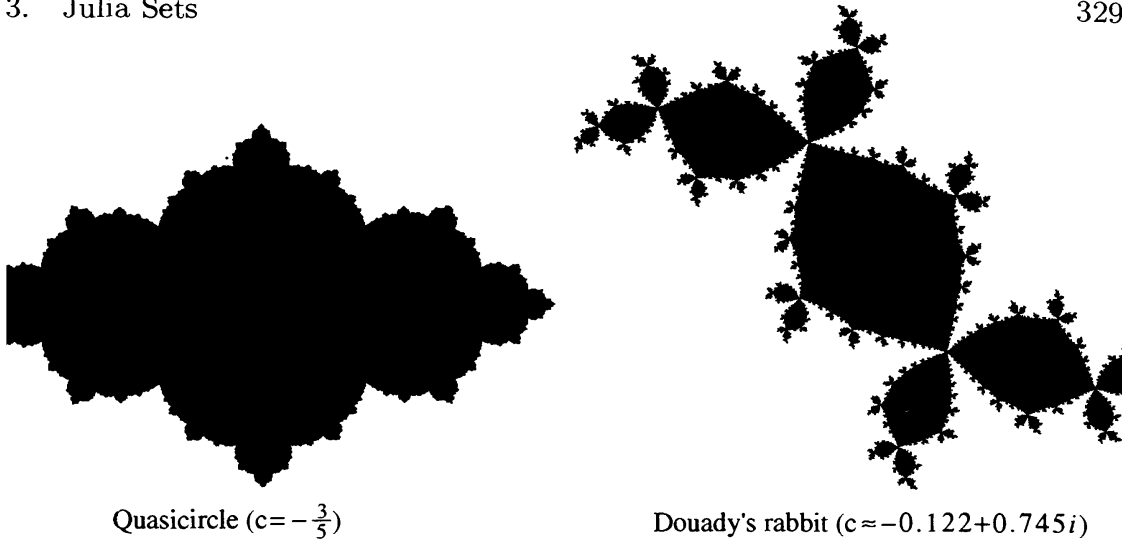


Cauliflower set ($c = \frac{1}{4}$)



Totally disconnected ($c = 0.251$)

The theorem above forms the basis for a method for producing computer images of Julia sets, called the **boundary scanning method**. Fix R large, fix N large, and fix a grid of points in the z -plane. For each point z in the grid, we compute the iterates $P^k(z)$ until we reach an integer $n = n(z)$ such that $|P^n(z)| > R$, or until we reach $k = N$, in which case we set $n(z) = N$.

Quasicircle ($c = -\frac{3}{5}$)Douady's rabbit ($c \approx -0.122 + 0.745i$)

If we color blue those points z for which $n(z) < N$ and we color green those points for which $n(z) = N$, we obtain a picture of the **filled-in Julia set**, which is the union of $\mathcal{J}$ and the bounded components of the Fatou set. The integer $n(z)$ measures the escape rate of z to ∞ . The escape rates can be captured visually by using different colors or shadings for different ranges of $n(z)$. An image of the Julia set can be obtained by recoloring red all those grid points for which $n(z) = N$ but that have a close neighbor satisfying $n(z) < N$.

Theorem. Let $P(z)$ be a polynomial of degree $d \geq 2$. If z_0 belongs to the Julia set $\mathcal{J}$ of $P(z)$ and if U is any open neighborhood of z_0 , then there exists $N \geq 1$ such that

$$(3.1) \quad \mathcal{J} \subseteq U \cup P(U) \cup \dots \cup P^N(U).$$

Further, the inverse iterates $\bigcup_{k=1}^{\infty} P^{-k}(z_1)$ of any point $z_1 \in \mathcal{J}$ are dense in $\mathcal{J}$.

The proof of this theorem depends in an essential way on the general version of Montel's theorem. By our hypothesis, the restrictions of the sequence of iterates $\{P^n(z)\}$ to U do not form a normal family. By Montel's theorem, there is at most one value that is omitted by the sequence $\{P^n(z)\}$ on U . If $\{P^n(z)\}$ does not omit any value, then

$$(3.2) \quad \bigcup_{n=1}^{\infty} P^n(U) \supseteq \mathcal{J}.$$

Since $\mathcal{J}$ is compact, and each set $P^n(U)$ is open, a finite number of these open sets already cover $\mathcal{J}$, and we obtain (3.1).

Suppose, on the other hand, there is an omitted value, call it w_0 , so that $\bigcup P^n(U) = \mathbb{C} \setminus \{w_0\}$. Then there is no $z_0 \neq w_0$ such that $P(z_0) = w_0$, or else we would have $z_0 \in P^m(U)$ for some m , and $w_0 \in P^{m+1}(U)$. Since the only solution of $P(z) = w_0$ is w_0 , $P(z) - w_0$ is a constant multiple of

$(z - w_0)^d$, and since $P(z)$ is assumed to be monic, $P(z) = w_0 + (z - w_0)^d$. Now, the iterates $P^n(z)$ tend uniformly to w_0 on any compact subset of $\{|z - w_0| < 1\}$. This can be seen directly, or by making the change of variable $\zeta = z - w_0$, which conjugates $P(z)$ to ζ^d . In any event, w_0 belongs to the Fatou set of $P(z)$, and we still obtain (3.2). Since $\mathcal{J}$ is compact, we then obtain (3.1) again. This proves the first statement of the theorem.

For the second statement, let U be an open nonempty subset of $\mathcal{J}$. Then $z_1 \in P^k(U)$ for some $k \geq 1$, and consequently, one of the points in $P^{-k}(z_1)$ belongs to U . This completes the proof.

This theorem provides another method for producing computer images of Julia sets, called the **inverse iteration method**. One selects a starting point $z_1 \in \mathcal{J}$, and one calculates the inverse iterates $P^{-k}(z_1)$ by solving the polynomial equations $P^k(z) = z_1$ for $1 \leq k \leq N$. For this method to be effective, the degree of $P(z)$ should be small (as $d = 2$). For a starting point in $\mathcal{J}$, one can look among the fixed points of $f(z)$. A rational function of degree $d \geq 2$ always has a fixed point in the Julia set (see Exercises 16 and 23).

The preceding theorem provides the basis for referring to Julia sets as “fractal sets,” in the sense that shapes in the Julia set reappear infinitely often at different scales near any point of the Julia set. The idea is as follows. Let $z_0 \in \mathcal{J}$ be such that the forward orbit of z_0 does not contain any of the critical points of $P(z)$, and let z_1 be any point of $\mathcal{J}$. By the theorem, we can find images $P^n(z_0)$ arbitrarily close to z_1 . If U is a small enough disk centered at z_0 , then $P^n(z)$ maps U conformally onto an open set containing $P^n(z_0)$. Since $\mathcal{J}$ is completely invariant, $P^n(z)$ maps $\mathcal{J} \cap U$ onto $\mathcal{J} \cap P^n(U)$. Hence $\mathcal{J} \cap P^n(U)$ has the same “shape” as $\mathcal{J} \cap U$ though at a different scale.

Exercises for XII.3

1. Show that the Julia set $\mathcal{J}$ and the Fatou set $\mathcal{F}$ of a rational function $f(z)$ satisfy $f(\mathcal{J}) = \mathcal{J}$ and $f(\mathcal{F}) = \mathcal{F}$.
2. Show that the rational function $f(z) = z^2/(z^2 + 1)$ is conjugate to the quadratic polynomial $P(z) = z^2 + 2$.
3. For $\lambda, \mu \neq 0$, the two maps $f(z) = \lambda z$ and $g(z) = \mu z$ of $\mathbb{C}^*$ are conjugate if and only if $\lambda = \mu$ or $\lambda = 1/\mu$. *Hint.* A conjugation maps fixed points to fixed points.
4. Let z_0 be a fixed point of $f(z)$. Define the **multiplier** of the fixed point z_0 to be $\lambda = f'(z_0)$. Show that the multiplier at a fixed point is a conjugation invariant, that is, if $\zeta = \varphi(z)$ conjugates $f(z)$ to $g(\zeta)$, then the multiplier of $g(\zeta)$ at the fixed point $\varphi(z_0)$ is equal to the multiplier of $f(z)$ at z_0 .

5. A fixed point z_0 of $f(z)$ is a **repelling fixed point** if $|f'(z_0)| > 1$. Show that the Julia set of a rational function $f(z)$ contains all its repelling fixed points.
6. Show that the Julia set of a fractional linear transformation is either empty or consists of one fixed point.
7. A fixed point z_0 of $f(z)$ is an **attracting fixed point** if $|f'(z_0)| < 1$. The **basin of attraction** of z_0 , denoted by $A(z_0)$, is the set of z whose iterates $f^n(z)$ converge to z_0 as $n \rightarrow \infty$. Show that if $f(z)$ is a rational function, then $A(z_0)$ is an open subset of $\mathbb{C}^*$ containing z_0 whose boundary coincides with the Julia set.
8. Show that if $f(z)$ is a rational function of degree d , then the m th iterate $f^m(z)$ is a rational function of degree d^m .
9. Show that the Julia set of a rational function $f(z)$ coincides with the Julia set of its m th iterate $f^m(z)$.
10. A point z_0 is a **periodic point** of $f(z)$ if it is a fixed point of $f^m(z)$ for some $m \geq 1$. For such a z_0 , set $z_1 = f(z_0)$, $z_2 = f(z_1)$, $\dots$, $z_{m-1} = f(z_{m-2})$. Show that each z_j is a fixed point of $f^m(z)$ with the same multiplier $\lambda = f'(z_0) \cdots f'(z_{m-1})$. *Remark.* Assuming that the z_j 's are distinct, we define the **multiplier of the cycle** $\{z_0, z_1, \dots, z_{m-1}\}$ to be the multiplier of $f^m(z)$ at any of the points of the cycle. The cycle is an **attracting cycle** if its multiplier λ satisfies $|\lambda| < 1$, and it is a **repelling cycle** if $|\lambda| > 1$. The integer m is the **period** of the periodic point, or the **length** of the cycle.
11. Find all repelling cycles of the polynomial $f(z) = z^2$.
12. Show that all repelling cycles of a rational function are contained in its Julia set.
13. Find all attracting cycles of length two of the quadratic polynomial $z^2 + c$. Show that the values of the complex parameter c for which there is an attracting cycle of length two form an open disk.
14. Let $f(z)$ be a rational function. We define the **basin of attraction of an attracting cycle** of $f(z)$ to consist of the points $z \in \mathbb{C}^*$ whose iterates $f^n(z)$ accumulate on the cycle as $n \rightarrow \infty$. Show that the basin of attraction of an attracting cycle is an open subset of $\mathbb{C}^*$ whose boundary coincides with the Julia set. Show that different points of an attracting cycle lie in different components of the basin of attraction.
15. We define the **multiplicity of a fixed point** z_0 of $f(z)$ to be the order of the zero of $f(z) - z$ at $z = z_0$. Show that a fixed point has

- multiplicity $m \geq 2$ if and only if its multiplier is 1. Show that the multiplicity of a fixed point is a conjugation invariant.
16. Show that a fixed point of a rational function $f(z)$ of multiplicity $m \geq 2$ belongs to the Julia set.
 17. Show that a rational function of degree d has $d + 1$ fixed points, counting multiplicity.
 18. Define the **analytic index** of a fixed point $z_0 \neq \infty$ of $f(z)$ to be the residue of $1/(z - f(z))$ at z_0 . (a) Show that if z_0 is a fixed point of $f(z)$ with multiplier $\lambda \neq 1$, then the analytic index of $f(z)$ at z_0 is $1/(1 - \lambda)$. (b) Show that if $f(z)$ has a fixed point of multiplicity m at z_0 , then for any small $\varepsilon > 0$, $f_\varepsilon(z) = f(z) - \varepsilon$ has m fixed points near z_0 , each of multiplicity one, for which the sum of the analytic indices of $f_\varepsilon(z)$ tends to the analytic index of $f(z)$ at z_0 as $\varepsilon \rightarrow 0$. (c) Show that the analytic index of a fixed point is a conjugation invariant.
 19. Find the fixed points and their analytic indices for the rational function $f(z) = (3z^2 + 1)/(z^2 + 3)$. Determine the Julia set and the Fatou set of $f(z)$.
 20. Suppose that $f(z) = z - z^{m+1} + Az^{2m+1} + \mathcal{O}(z^{2m+2})$ for some integer $m \geq 1$. Show that the analytic index of $f(z)$ at the fixed point $z = 0$ is A .
 21. Suppose that $g(z) = z - z^{m+1} + \mathcal{O}(z^{m+2})$ for some integer $m \geq 1$. Show that $g(z)$ can be conjugated near 0 to $f(z) = z - z^{m+1} + Az^{2m+1} + \mathcal{O}(z^{2m+2})$, where A is the analytic index of $f(z)$ at $z = 0$. *Hint.* Try conjugating by $\varphi(z) = z(1 + az^k)$.
 22. If ∞ is a fixed point for $f(z)$, we define the **analytic index at ∞** of $f(z)$ to be the analytic index of $1/f(1/\zeta)$ at $\zeta = 0$. (*Danger!* This is not the residue of $z - f(z)$ at $z = \infty$.) Show that the sum of the analytic indices of a rational function $f(z)$ at its fixed points in $\mathbb{C}^*$ is $+1$.
 23. Show that if $f(z)$ is a rational function of degree $d \geq 2$, and if all the fixed points of $f(z)$ have multiplicity one, then $f(z)$ has at least one repelling fixed point. *Hint.* Use Exercise 18a, and sum the real parts of the analytic indices at the $d + 1$ fixed points.
 24. Let $f(z)$ be a rational function of degree $d \geq 2$. (a) Show that $\mathcal{J}$ is nonempty. (b) Show that either $\mathcal{F}$ is empty or $\mathcal{F}$ is dense in $\mathbb{C}^*$. (c) Show that $\mathcal{J}$ has no isolated points.

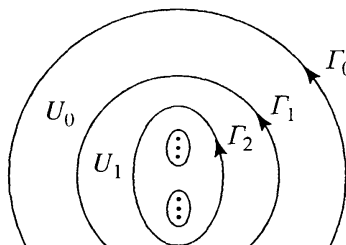
25. Let $f(z)$ be a rational function of degree $d \geq 2$. Show that the repelling periodic points of $f(z)$ are dense in the Julia set of $f(z)$. *Remark.* This theorem was proved independently by Fatou and Julia. It can be regarded as the first substantial theorem in rational iteration theory. For the proof, fill in the details of the following argument. Using the fact that $\mathcal{J}$ is a compact set with no isolated points, show that the points $z_0 \in \mathcal{J}$ that are not in the forward orbit of a critical point are dense in $\mathcal{J}$. For such a point z_0 , refer to the entire function $g(\zeta)$ from Exercise 2.11. Show that there is a point $\zeta_0 \in \mathbb{C}$ such that ζ_0 is not a critical point of $g(\zeta)$ and $w_0 = g(\zeta_0)$ satisfies $f^m(w_0) = z_0$ for some m . Show that the points η_n from Exercise 2.11 are repelling periodic points of $f(z)$ that converge to z_0 .

4. Connectedness of Julia Sets

The critical points and their forward orbits play a special role in complex dynamics. In this section we prove two results on connectedness of Julia sets, in which the critical points play the crucial role. While we focus on polynomials, the main ideas of the proofs are flexible and can be adapted to other situations (see the exercises).

We consider a monic polynomial $P(z) = z^d + \mathcal{O}(z^{d-1})$ of degree $d \geq 2$. Let $R > 0$ be large, so that the circle $\Gamma_0 = \{|z| = R\}$ is contained in the basin of attraction $A(\infty)$ of ∞ , and $|P(z)| > R$ for $|z| \geq R$. We may assume that $\Gamma_1 = P^{-1}(\Gamma_0)$ is “almost” a circle of radius $R^{1/d}$, which is mapped d -to-one by $P(z)$ onto Γ_0 . Let U_0 be the “annular” domain between Γ_0 and Γ_1 , and define $\Gamma_k = P^{-k}(\Gamma_0)$ and $U_k = P^{-k}(U_0)$. Then each U_k is an open set with boundary $\Gamma_k \cup \Gamma_{k+1}$, and $P(z)$ maps U_k d -to-one onto U_{k-1} (counting multiplicity at critical points).

If $z \in A(\infty)$, then either $|z| > R$, or there is a first integer $k \geq 0$ such that $|P^{k+1}(z)| > R$. In the latter case, $P^k(z)$ belongs to $U_0 \cup \Gamma_0$, and $z \in U_k \cup \Gamma_k$. Thus $A(\infty)$ is the disjoint union of the exterior disk $\{|z| > R\}$, the open sets U_k , and the curves Γ_k separating U_k and U_{k-1} . The effect of P is to map each Γ_k to Γ_{k-1} and each U_k to U_{k-1} .



Since there are only finitely many critical points, we can arrange by adjusting R that no critical point of $P(z)$ lies on a curve Γ_k . Then $P(z)$ is conformal at each point of Γ_k and maps Γ_k d -to-one onto Γ_{k-1} . The composition $P^k(z)$ is then conformal at each point of Γ_k and maps Γ_k d^k -to-one onto the circle Γ_0 . Thus each Γ_k is a finite union of disjoint simple closed analytic curves. The Γ_k 's accumulate on $\partial A(\infty)$ as $k \rightarrow \infty$, and since $\mathcal{J} = \partial A(\infty)$, we have

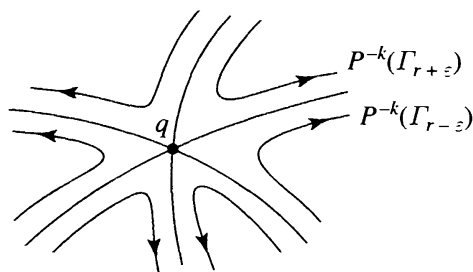
$$\mathcal{J} = \lim_{k \rightarrow \infty} \Gamma_k = \lim_{k \rightarrow \infty} U_k.$$

Let $W_k = P^{-k}(\{|z| < R\})$. The boundary of W_k is Γ_k , and W_k is obtained from the disk $\{|z| < R\}$ by removing $U_0 \cup \Gamma_1 \cup \cdots \cup U_{k-1} \cup \Gamma_k$. Since the complement of W_k is connected, each component of W_k is simply connected. In view of the Riemann mapping theorem, we can think of W_k as the union of a finite number of "analytic disks" with analytic boundaries, one domain for each closed curve in Γ_k . Since $\mathbb{C} \setminus W_k \subset A(\infty)$, W_k contains all the bounded components of the Fatou set $\mathcal{F}$ together with the Julia set $\mathcal{J}$. Note also that $P^{-1}(W_k) = W_{k+1} \subset W_k$. Now the stage is set, and we state and prove our two theorems.

Theorem. *Let $P(z)$ be a polynomial of degree $d \geq 2$. The Julia set $\mathcal{J}$ of $P(z)$ is connected if and only if the iterates of each critical point of $P(z)$ are bounded.*

Suppose first that the iterates of critical points are all bounded, so that no critical point of $P(z)$ is in $A(\infty)$. Then at each point $w = P(z) \in A(\infty)$ we can define d analytic branches of the inverse function $P^{-1}(w)$, and these branches can be continued analytically along any curve in $A(\infty)$. Similarly, there are d^k analytic branches of the inverse $P^{-k}(w)$ of the k th iterate $P^k(z)$, and $P^{-k}(w)$ can be continued analytically along any path in $A(\infty)$. We deform Γ_0 to Γ_1 through a family of circular closed disjoint curves Γ_t in U_0 , $0 \leq t \leq 1$. (In fact, since U_0 is conformally equivalent to an annulus, we can arrange for the Γ_t 's to be simple closed analytic curves.) Suppose Γ_k consists of just one closed curve. Then as w traverses the closed curve Γ_0 d^k times, an analytic branch of $P^{-k}(w)$ traverses Γ_k once and returns to its initial point. We deform the curve Γ_0 through the Γ_t 's to Γ_1 , and we follow a branch of $P^{-k}(w)$ d^k times around each Γ_t . Since there are no critical points in $A(\infty)$, the branch of $P^{-k}(w)$ moves continuously with the curve Γ_t and cannot return to the initial value until w completes d^k full loops around Γ_t . Hence each $P^{-k}(\Gamma_t)$ consists of a single closed curve, and in particular, $\Gamma_{k+1} = P^{-k}(\Gamma_1)$ consists of a single closed curve. By induction on k we see, then, that each Γ_n consists of one closed curve, and in particular, each Γ_n is connected. Since a limit of compact connected sets is connected (Exercise 7), and since $\mathcal{J}$ is the limit of the connected sets Γ_n , $\mathcal{J}$ is connected.

To prove the converse, suppose next that there is a critical point in $A(\infty)$. Let k be the first integer such that there is a critical point in U_k , and for simplicity assume that there is only one critical point q in U_k . The argument in the preceding paragraph shows that each Γ_j consists of just one curve for $j \leq k$. As before, we consider the circular curves Γ_t and their inverse images $P^{-k}(\Gamma_t)$. Let r be the smallest parameter value such that $q \in P^{-k}(\Gamma_r)$. We can assume that Γ_t is a circle for t near r . Fix $\varepsilon > 0$ small. The argument in the preceding paragraph shows that $P^{-k}(\Gamma_{r-\varepsilon})$ consists of just one curve, which is the boundary of a bounded simply connected domain V containing q . Consider the inverse image $P^{-k}(\Gamma_r)$, which includes the critical point of q . If the critical point has order m , then the part of $P^{-k}(\Gamma_r)$ near q consists of $2m + 2$ analytic arcs that terminate at q and divide a disk centered at q into $2m + 2$ sector-like domains, as in the figure. In the sectors we alternately have $P^k(z)$ outside Γ_r and inside Γ_r . The curve $P^{-k}(\Gamma_{r-\varepsilon}) = \partial V$ enters and then exits every other sector, as in the figure. The curves in $P^{-k}(\Gamma_{r+\varepsilon})$ enter and then exit each of the remaining sectors. We can think of V as a disk, via the Riemann mapping theorem. Line segments drawn from q to the nearest point in ∂V in each of the $m + 1$ sectors where ∂V approaches q then divide V into $m + 1$ components. Since each of these components contains an arc of $P^{-k}(\Gamma_{r+\varepsilon})$, we see that $P^{-k}(\Gamma_{r+\varepsilon})$ consists of at least $m + 1$ analytic curves. Hence W_{k+1} has at least $m + 1$ components. Since each of these components contains curves in Γ_j for all $j > k + 1$, each of these components also contains points of $\mathcal{J}$, and the Julia set is not connected.

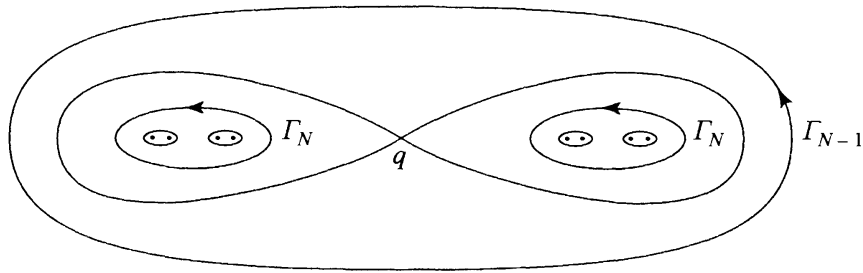


critical point of order $m = 2$

We say that a compact subset K of $\mathbb{C}$ is **totally disconnected** if for every $z_0 \in K$ and $\varepsilon > 0$ there is a subset E of K such that $z_0 \in E$, the diameter of E is less than ε , and E is at a positive distance from $K \setminus E$. Thus $K = E \cup (K \setminus E)$ is a decomposition of K into two disjoint compact subsets, one of which contains z_0 and has small diameter. At the opposite end of the spectrum from the preceding theorem, we now have the following.

Theorem. *Let $P(z)$ be a polynomial of degree $d \geq 2$. If all the critical points of $P(z)$ are iterated to ∞ , then the Julia set $\mathcal{J}$ of $P(z)$ is totally disconnected, and the Fatou set $\mathcal{F}$ of $P(z)$ coincides with the basin of attraction $A(\infty)$ of ∞ .*

In this case we choose N so large that none of the critical points of $P(z)$ are in W_N . Then since each component of W_N is simply connected, all d branches of $P^{-1}(w)$ can be defined analytically on each component of W_N . Each branch has image in W_N and maps points of $\mathcal{J}$ to points of $\mathcal{J}$. Fix $z_0 \in \mathcal{J}$. Let V be a component of W_N that contains $P^k(z_0)$ for infinitely many k 's. For such a k , let $g_k(w)$ be the branch of $P^{-k}(w)$ on V satisfying $g_k(P^k(z_0)) = z_0$. The family $\{g_k(w)\}$ is uniformly bounded on V , hence a normal family of analytic functions on the open set V . Moreover, $g_k(w) \in W_{N+k} \cap A(\infty)$, so $g_k(w) \rightarrow \mathcal{J}$ for all $w \in W_N \cap A(\infty)$. Consequently, any limit $g(w)$ of the $g_k(w)$'s as $k \rightarrow \infty$ maps $W_N \cap A(\infty)$ into $\mathcal{J}$. Since $\mathcal{J}$ has no interior points, $g(w)$ is constant, and the constant value of $g(w)$ must be z_0 . Thus there is a subsequence k_j for which $g_{k_j}(w)$ is defined and converges to z_0 uniformly on compact subsets of V . If we shrink V slightly, we obtain a compact subset E_0 of V such that $P^{k_j}(z_0) \in E_0$, $\partial E_0 \subset A(\infty)$, and the g_{k_j} 's converge uniformly on E_0 to z_0 . Set $E_j = g_{k_j}(E_0)$. Then E_j is compact, $z_0 = g_{k_j}(P^{k_j}(z_0)) \in E_j$, and the diameter of E_j tends to 0. Since $\partial E_0 \subset A(\infty)$, also $\partial E_j \subset A(\infty)$. Thus $E_j \cap \mathcal{J}$ has positive distance from $\mathcal{J} \setminus E_j$, and the conditions in the definition of total disconnectedness are fulfilled at z_0 . Since $z_0 \in \mathcal{J}$ is arbitrary, the Julia set is totally disconnected.



Exercises for XII.4

1. For each of the quadratic polynomials $P(z) = z^2$ and $P(z) = z^2 + 2$, sketch roughly the curves Γ_k for $0 \leq k \leq 3$. Take Γ_0 to be the circle $\{|z| = 9\}$.
2. Let $P(z)$ be a polynomial of degree $d \geq 2$. Let $G(z)$ be the Riemann map of $A(\infty)$ onto the exterior of the unit circle $\{|\zeta| > 1\}$ such that $G(\infty) = \infty$. Show that $\zeta = G(z)$ conjugates the polynomial $P(z)$ on $A(\infty)$ to the map ζ^d on the exterior $\{|\zeta| > 1\}$ of the unit circle. *Hint.* Apply the Schwarz reflection principle to $f = G \circ P \circ G^{-1}$.

3. Let $f(z)$ be a rational function with an attracting fixed point at z_0 . Let U be the connected component of the basin of attraction of z_0 (see Exercise 3.7) containing z_0 . Suppose there is no critical point of $f(z)$ in U except for possibly z_0 . (a) Show that U is simply connected. *Hint.* Conjugate z_0 to ∞ by a fractional linear transformation and assume $z_0 = \infty$. Then modify the proof in the text. (b) Show that if $f(z)$ has degree $d \geq 2$, then U is conformally equivalent to the open unit disk, and $f(z)$ has a critical point at z_0 . (c) Show that if z_0 is a critical point of $f(z)$ of order $m \geq 1$, and if $G(z)$ is the Riemann map of U onto the unit disk $\mathbb{D}$ such that $G(z_0) = 0$, then $\zeta = G(z)$ conjugates $f(z)$ on U to the map ζ^m on $\mathbb{D}$.
4. Let $f(z)$ be a rational function of degree $d \geq 2$, and suppose $\{z_0, f(z_0), \dots, f^{n-1}(z_0)\}$ is an attracting cycle for $f(z)$. Show that $f(z)$ has a critical point in one of the connected components of the basin of attraction of the cycle that contains a point of the cycle. *Hint.* Apply the preceding exercise to $f^n(z)$. For definitions, see Exercises 3.10 and 3.14.
5. Show that a rational function of degree $d \geq 2$ has at most $2d - 2$ attracting cycles. *Hint.* Count critical points and apply the preceding exercise.
6. Show that a polynomial $P(z)$ of degree $d \geq 2$ has at most $d - 1$ attracting cycles in the (finite) complex plane.
7. Let $\{E_n\}$ be a sequence of compact subsets of $\mathbb{C}$ that converges to a compact set E . Thus E consists of those points z for which there are $z_n \in E_n$ satisfying $z_n \rightarrow z$. Show that if each E_n is connected, then E is connected.
8. Show that a compact subset K of $\mathbb{C}$ is totally disconnected if and only if for each $z_0 \in K$ and $\varepsilon > 0$, there is a simple closed curve γ in $\mathbb{C} \setminus K$ of diameter less than ε such that γ separates z_0 from ∞ , that is, z_0 lies in the bounded component of $\mathbb{C} \setminus \gamma$.
9. Show that a compact subset K of $\mathbb{C}$ is totally disconnected if and only if K does not contain a continuum. *Strategy.* For fixed $z_0 \in K$, let $E(z_0)$ be the intersection of all closed subsets E of K such that $z_0 \in E$ and $K \setminus E$ is closed. Show that $E(z_0)$ is either a continuum or the singleton $\{z_0\}$. The set $E(z_0)$ is the “connected component of z_0 in K .”
10. Let z_0 be a fixed point of a polynomial $P(z)$. Show that either $\{z_0\}$ is a connected component of the Julia set $\mathcal{J}$, or else the connected component of z_0 in the filled-in Julia set $\mathcal{K} = \mathbb{C} \setminus A(\infty)$ contains a critical point of $P(z)$.

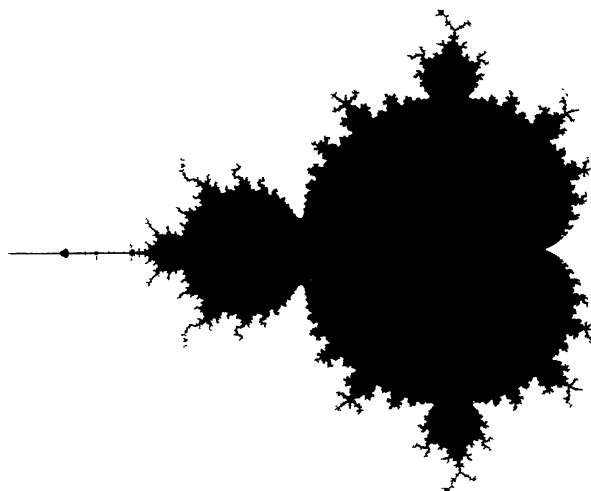
5. The Mandelbrot Set

We now turn our attention to some special polynomials, the quadratic polynomials

$$P_c(z) = z^2 + c, \quad z \in \mathbb{C},$$

where $c \in \mathbb{C}$ is a complex parameter. Let $\mathcal{J}_c$ denote the Julia set of $P_c(z)$. The only critical point of $P_c(z)$ is the origin $z = 0$. We consider the iterates $P_c(0) = c$, $P_c^2(0) = c^2 + c$, $P_c^3(0) = (c^2 + c)^2 + c$, There are two cases that can occur, corresponding to the two theorems in the preceding section. If the iterates $P_c^k(0)$ are bounded, then the Julia set $\mathcal{J}_c$ is connected. Otherwise, the iterates $P_c^k(0)$ tend to ∞ , and then $\mathcal{J}_c$ is totally disconnected.

We define the **Mandelbrot set** $\mathcal{M}$ to be the set of parameter values c such that the iterates $P_c^k(0)$ are bounded. Thus $c \in \mathcal{M}$ if and only if $\mathcal{J}_c$ is connected. Note that the Mandelbrot set is a subset of parameter space (c -space) and not of dynamic space (z -space).



Theorem. *A complex number c belongs to the Mandelbrot set $\mathcal{M}$ if and only if $|P_c^n(0)| \leq 2$ for all $n \geq 1$. The Mandelbrot set is a compact subset of the closed disk $\{|c| \leq 2\}$. Further, $\mathbb{C} \setminus \mathcal{M}$ is connected.*

Suppose that $|c| > 2$. If $|z| = |c|$, then $|z^2 + c| \geq |c|^2 - |c| = (|c| - 1)|z|$. By the maximum principle, applied to $z/(z^2 + c)$ on the exterior of the circle, this estimate $|P_c(z)| \geq (|c| - 1)|z|$ persists for all z satisfying $|z| \geq |c|$. By iterating, starting with $P_c(0) = c$, we obtain successively $|P_c^2(0)| \geq (|c| - 1)|c|$, $|P_c^3(0)| \geq (|c| - 1)^2|c|$, and eventually $|P_c^k(0)| \geq (|c| - 1)^{k-1}|c|$. Thus $P_c^k(0) \rightarrow \infty$, so $c \notin \mathcal{M}$. It follows that $\mathcal{M}$ is a subset of the closed disk $\{|c| \leq 2\}$.

Suppose next that $|c| \leq 2$ and that $c \notin \mathcal{M}$. Let $n \geq 1$ be the first integer such that $|P_c^n(0)| > 2$. Then $|P_c^{n+1}(0)| = |P_c^n(0)^2 + c| \geq |P_c^n(0)|^2 - |c| > (|P_c^n(0)| - 1)|P_c^n(0)|$. We estimate $P_c^{n+2}(0)$ in the same way, and so on,

and we obtain $|P_c^{n+k}(0)| > (|P_c^n(0)| - 1)^k |P_c^n(0)|$, which tends to $+\infty$ as $k \rightarrow \infty$. This proves the first statement in the theorem.

For fixed n , the set of $c \in \mathbb{C}$ such that $|P_c^n(0)| \leq 2$ is a closed set. Thus the description of $\mathcal{M}$ as the set of $c \in \mathbb{C}$ such that $|P_c^n(0)| \leq 2$ for all n shows that $\mathcal{M}$ is closed, and it is bounded; hence $\mathcal{M}$ is compact. Let U be any bounded domain with boundary contained in $\mathcal{M}$. Then $|P_c^n(0)| \leq 2$ for all $c \in \partial U$. Since $P_c(0)$ is a polynomial in c , we infer from the maximum principle that $|P_c^n(0)| \leq 2$ for all $c \in U$. Hence $U \subset \mathcal{M}$. It follows that $\mathbb{C} \setminus \mathcal{M}$ has no bounded components, and so $\mathbb{C} \setminus \mathcal{M}$ consists of a single unbounded component. The theorem is proved.

The constant 2 in the theorem is sharp. For $c = -2$, the origin is iterated in two steps to the repelling fixed point $z_0 = 2$, $0 \rightarrow -2 \rightarrow +2 \rightarrow +2 \rightarrow \dots$. In particular, the iterates $P_{-2}^k(0)$ are bounded, so $-2 \in \mathcal{M}$. It is straightforward to show (Exercise 3) that $\mathcal{M}$ meets the real axis in the closed interval $[-2, \frac{1}{4}]$.

The theorem suggests an algorithm for obtaining computer pictures of $\mathcal{M}$. We fix a large integer N , and we lay a grid of points on the disk $\{|c| \leq 2\}$. For each point c in the grid we compute the iterates $P_c^k(0)$ of 0. If we reach $k < N$ for which $|P_c^k(0)| > 4$, we color the grid point sky blue; otherwise, we stop when we reach $k = N$ and we color the grid point black. The Mandelbrot set is approximated by the black grid points. By using different shades of blue to depict different escape rates, we obtain more attractive pictures. For an assortment of close-up pictures of various parts of the Mandelbrot set and a discussion of algorithms for obtaining the pictures, see the award-winning book of Peitgen and Richter [PR].

We describe now some results on the Mandelbrot set without giving proofs. Several of the proofs are laid out as exercises, while some are beyond our scope. We focus on the set W of those parameter values c such that $P_c(z)$ has an attracting cycle (other than ∞). Recall from the exercises for Sections 3 and 4 the definitions of attracting fixed points, attracting cycles, and their basins of attraction.

Suppose $P_c(z)$ has an attracting cycle, other than the fixed point at ∞ . The basin of attraction of the cycle is contained in the Fatou set $\mathcal{F}_c$ and is disjoint from $A(\infty)$. Hence $\mathcal{J}_c$ cannot be totally disconnected, and $c \in \mathcal{M}$. Thus $W \subset \mathcal{M}$. It is straightforward to check that W is an open subset of the complex plane (Exercise 8). This depends on the observation that an attracting periodic point of period n is a zero of $P_c^n(z) - z$, and by Hurwitz's theorem the zeros of this polynomial vary continuously with the parameter c . With somewhat more effort it can be proved that $\partial W \subseteq \partial \mathcal{M}$, so that W is a union of connected components of the interior of $\mathcal{M}$ (Exercise 10). These components are called the **hyperbolic components** of the interior of $\mathcal{M}$. Each hyperbolic component corresponds to a single attracting cycle $\{z_0(c), z_1(c), \dots, z_{n-1}(c)\}$ of length n , where each n -periodic point $z_j(c)$ is an analytic function of c in the hyperbolic component.

It turns out (see [CG]) that if U is a hyperbolic component of the interior of $\mathcal{M}$, and if $\lambda(c)$ is the multiplier of the attracting cycle of $P_c(z)$, then $c \mapsto \lambda(c)$ maps U conformally onto the open unit disk in the complex plane. In particular, each of the hyperbolic components contains exactly one parameter value c_0 such that $P_{c_0}(z)$ has a cycle with multiplier 0, that is, a “superattracting cycle.” We refer to c_0 as the **center of the hyperbolic component**. The centers of the hyperbolic components corresponding to superattracting cycles of length n can be calculated by solving $P_c^n(0) = 0$, which is a polynomial equation in c , and discarding the solutions of $P_c^m(0) = 0$ for $m < n$.

Example. The c ’s for which $P_c(z)$ has an attracting fixed point form a cardioid, called the **principal cardioid**. It is a hyperbolic component of the interior of $\mathcal{M}$, with center at $c = 0$ corresponding to the superattracting fixed point $z = 0$ for $P_0(z) = z^2$. (See Exercise 4.)

Example. To find the superattracting 2-cycles, we solve $P_c^2(0) = c^2 + c = 0$. The solution $c = 0$ corresponds to the superattracting fixed point $z = 0$ for $P_0(z) = z^2$. The other solution $c = -1$ corresponds to the superattracting 2-cycle $\{0, -1\}$ for $P_{-1}(z) = z^2 - 1$. The corresponding hyperbolic component of the interior of $\mathcal{M}$ is the disk $\{|c + 1| < \frac{1}{4}\}$. (See Exercise 3.13.)

It was known for some time that the Mandelbrot set $\mathcal{M}$ is connected, that is, $\mathbb{C}^* \setminus \mathcal{M}$ is simply connected. Some intense interest has been focused on the following two questions. Does the conformal map from the open unit disk to $\mathbb{C}^* \setminus \mathcal{M}$ extend continuously to the boundary of the disk? Are all components of the interior of $\mathcal{M}$ hyperbolic? The answers remain a mystery. For more on the Mandelbrot set, see [CG].

Exercises for XII.5

1. Show that if $|P_c^n(0)| > 2$, then $|P_c^k(0)|$ is strictly increasing for $k \geq n$.
2. Show that if $|c| \leq 2$, then the Julia set $\mathcal{J}_c$ is a subset of the closed disk $\{|z| \leq 2\}$.
3. Show that $\mathcal{M} \cap \mathbb{R} = [-2, \frac{1}{4}]$. *Hint.* If $-2 \leq c \leq \frac{1}{4}$, let $r = (1 + \sqrt{1 - 4c})/2$, the larger root of $x^2 + c = x$. Show that $P_c([c, r]) \subseteq [c, r]$.
4. Show that $P_c(z)$ has an attracting fixed point with multiplier λ , $|\lambda| < 1$, if and only if $c = \lambda/2 - \lambda^2/4$. Show that these values c form a cardioid with cusp at $c = \frac{1}{4}$. Sketch the cardioid. Show that the map $c \mapsto \lambda(c)$ is the Riemann map of the cardioid onto the open unit disk.

5. Find the values c for which there are superattracting cycles of length 3 and of length 4. Locate the values of c on the picture of the Mandelbrot set.
6. Show that any quadratic polynomial $P(z) = Az^2 + Bz + C$, $A \neq 0$, is conjugate to the polynomial $P_c(z)$ for a unique value of c .
7. Show that any quadratic polynomial $P(z) = Az^2 + Bz + C$ is conjugate to the polynomial $Q_\lambda(z) = \lambda z + z^2$ for some complex number λ . Show that $Q_\lambda(z)$ is conjugate to $Q_\mu(z)$ if and only if $\lambda = \mu$ or $\lambda = 2 - \mu$.
8. Show that the set of parameter values c such that $P_c(z)$ has an attracting cycle is an open subset of the complex plane that is contained in $\mathcal{M}$.
9. Use the result of Exercise 4.4 to show that if $P_c(z)$ has an attracting cycle of length n , then $P_c^{kn}(0)$ converges to a point $z_0(c)$ of the cycle as $k \rightarrow \infty$. Show further that $P_c^{kn+j}(0)$ converges to the point $z_j(c) = P_c(z_0(c))$ of the cycle as $k \rightarrow \infty$, and that the points $z_j(c)$ depend analytically on c , $0 \leq j < n - 1$.
10. Suppose $P_{c_0}(z)$ has an attracting cycle of length n . Let W be the connected component of the interior of $\mathcal{M}$ containing c_0 . Show that $P_c(z)$ has an attracting cycle of length n for each $c \in W$. *Hint.* Use the fact that $P_c^k(0)$ is uniformly bounded for $c \in W$, hence a normal family. Use the preceding exercise, and show that any limit of a subsequence is an attracting periodic point.

XIII

Approximation Theorems

In this chapter we prove two fundamental theorems, one “additive” and the other “multiplicative,” on prescribing zeros, and poles of meromorphic functions. The first is the Mittag-Leffler theorem, which asserts that we can prescribe the poles and principal parts of a meromorphic function. The second is the Weierstrass product theorem, which asserts that we can prescribe the zeros and poles, including orders, of a meromorphic function. The theorems are closely related. Both theorems are proved by the same type of approximation procedure, which depends on Runge’s theorem on approximation by rational functions. We prove Runge’s theorem in Section 1, followed by the Mittag-Leffler theorem in Section 2. In Section 3 we introduce infinite products, which can always be converted to infinite series by taking logarithms. We prove the Weierstrass product theorem in Section 4.

1. Runge’s Theorem

Suppose $f(z)$ is a complex-valued function on a compact subset K of the complex plane. When can $f(z)$ be approximated uniformly on K by rational functions? When can $f(z)$ be approximated uniformly on K by polynomials (in z)? Any uniform limit of rational functions with poles off K must be continuous on K , and further, since a uniform limit of analytic functions is analytic, it must be analytic on the interior of K . These are necessary conditions for approximation. The first nontrivial sufficient condition for approximation is given by the following theorem.

Theorem (Runge’s Theorem). *Let K be a compact subset of the complex plane. If $f(z)$ is analytic on an open set containing K , then $f(z)$ can be approximated uniformly on K by rational functions with poles off K .*

For the proof, let D be an open set with piecewise smooth boundary such that $K \subset D$ and such that $f(z)$ is analytic on $D \cup \partial D$. By the Cauchy

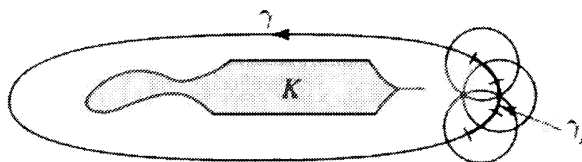
integral formula,

$$f(z) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad z \in D.$$

We chop ∂D up into a union of short curves γ_j such that each γ_j is contained in a disk $\{|z - c_j| < r_j\}$ that is at a positive distance from K . Then $f(z) = \sum f_j(z)$, where

$$f_j(z) = \frac{1}{2\pi i} \int_{\gamma_j} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad z \notin \gamma_j.$$

The function $f_j(z)$ is analytic off γ_j (Morera's theorem) and vanishes at ∞ . Each $f_j(z)$ has a Laurent expansion in descending powers of $z - c_j$ that converges uniformly for $|z - c_j| > r_j$, hence uniformly on K . Thus $f_j(z)$ is uniformly approximable on K by polynomials in $1/(z - c_j)$. Adding these approximants, we see that $f(z)$ is uniformly approximable on K by rational functions with poles off K .



There is some flexibility with respect to the location of the poles of the approximating rational functions.

Lemma. *Let K be a compact subset of the complex plane, let U be a connected open subset of the extended complex plane $\mathbb{C}^*$ disjoint from K , and let $z_0 \in U$. Every rational function with poles in U can be approximated uniformly on K by rational functions with poles at z_0 .*

Here we consider a polynomial in z to be a rational function with pole at ∞ . The lemma is established by a “translation of poles” argument. We define the set V to consist of those points $\zeta \in U$ such that $1/(z - \zeta)$ is approximable uniformly on K by rational functions with pole at z_0 . Thus if $\zeta \in V$, then each power $1/(z - \zeta)^k$ is also uniformly approximable on K by rational functions with pole at z_0 . Since $z_0 \in V$, the set V is nonempty. Since U is connected, to show that $V = U$, it suffices to show that V is open and closed in U . The “closed” assertion is easy. If $\zeta \in U$ is a limit of a sequence $\zeta_j \in V$, then $1/(z - \zeta_j)$ converges uniformly on K to $1/(z - \zeta)$, so by the definition of V , also $\zeta \in V$. The crux of the proof then is to show that V is open, and this reduces to showing that if $\zeta_0 \in V$, and if ζ is near ζ_0 , then $1/(z - \zeta)$ is uniformly approximable on K by polynomials in $1/(z - \zeta_0)$. We begin with the special case $\zeta_0 = \infty$. We must show that if ζ is near ∞ (that is, $|\zeta|$ is large), then $1/(z - \zeta)$ is uniformly approximable

on K by polynomials in z . For this, we expand $1/(z - \zeta)$ in a geometric series,

$$\frac{1}{z - \zeta} = -\frac{1}{\zeta} \frac{1}{1 - z/\zeta} = -\frac{1}{\zeta} \sum_{k=0}^{\infty} \frac{z^k}{\zeta^k}.$$

If $|z| \leq C$ for all $z \in K$, and if $|\zeta| > 2C$, then the k th term of the series is dominated by $1/2^k$. By the Weierstrass M -test, the series converges uniformly on K , and consequently, $\zeta \in V$ whenever $|\zeta| > 2C$. If ζ_0 is finite, the proof is essentially the same, with a change of variable $z \mapsto 1/(z - \zeta_0)$ to place ζ_0 at ∞ . We choose $\varepsilon > 0$ less than the distance from ζ_0 to K , so that $|z - \zeta_0| \geq \varepsilon$ for $z \in K$. If $|\zeta - \zeta_0| < \varepsilon/2$, then $|(\zeta - \zeta_0)^k / (z - \zeta_0)^k| < 1/2^k$ for $z \in K$, so the geometric series

$$\frac{1}{z - \zeta} = \frac{1}{z - \zeta_0} \frac{1}{1 - (\zeta - \zeta_0)/(z - \zeta_0)} = \frac{1}{z - \zeta_0} \sum_{k=0}^{\infty} \frac{(\zeta - \zeta_0)^k}{(z - \zeta_0)^k}$$

is uniformly convergent for $z \in K$, by the Weierstrass M -test. Thus all ζ satisfying $|\zeta - \zeta_0| < \varepsilon/2$ belong to V , and V is an open set. This proves the lemma.

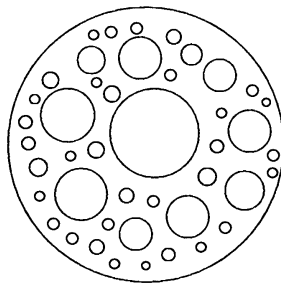
By approximating with rational functions and then using the lemma to translate the poles, we obtain immediately the following sharper version of Runge's theorem.

Theorem. *Let K be a compact subset of the complex plane, and suppose that $f(z)$ is analytic on an open set containing K . Let S be a subset of $\mathbb{C}^* \setminus K$ such that each connected component of $\mathbb{C}^* \setminus K$ contains a point of S . Then $f(z)$ can be approximated uniformly on K by rational functions with poles in S .*

In particular, if K is a compact subset of the complex plane, and if the complement of K is connected, then each function analytic in a neighborhood of K can be approximated uniformly on K by polynomials in z . There is a considerably more difficult theorem on polynomial approximation, **Mergelyan's theorem**, which asserts that any function that is continuous on K and analytic on the interior of K can be approximated uniformly on K by polynomials in z . The analogous statement for rational approximation is false.

Example. There is a compact set K with empty interior, and a continuous function $f(z)$ on K , such that $f(z)$ cannot be approximated uniformly on K by rational functions. We outline the construction of such an example, known as a **Swiss cheese set**. Let S be a sequence that is dense in the open unit disk $\mathbb{D}$. Select disks $D_k = \{|z - c_k| < r_k\}$ by induction, such that the center c_k of D_k is the first point in the sequence S that is not contained in any of the closed disks $E_j = \{|z - c_j| \leq r_j\}$ for $j < k$, and then choose

r_k so small that E_k is disjoint from the sets E_j for $j < k$. We also assume that $\sum r_k$ converges and $\sum r_k^2 < 1$. Define $K = \{|z| \leq 1\} \setminus \bigcup_{j=1}^{\infty} \Delta_j$, a compact set with no interior points. If $f(z)$ is a continuous function on K such that $|f(z)| \leq M$, then $|\int_{\partial D_k} f(z) dz| \leq 2\pi M r_k$, so that the series $\sum \int_{\partial D_k} f(z) dz$ converges absolutely. Further, $\sum \int_{\partial D_k} f(z) dz = \int_{\partial \mathbb{D}} f(z) dz$ if $f(z)$ is a rational function with poles off K , hence whenever $f(z)$ is any uniform limit on K of rational functions. However, by Green's theorem, or by a direct calculation, $\int_{\partial D_k} \bar{z} dz = 2\pi r_k^2 i$ and $\int_{\partial \mathbb{D}} \bar{z} dz = 2\pi i$. Since $\sum \int_{\partial D_k} \bar{z} dz \neq \int_{\partial \mathbb{D}} \bar{z} dz$, the function $\bar{z}$ cannot be approximated uniformly on K by rational functions.



Swiss cheese set

There are many applications of Runge's theorem to construct functions that do one thing or another. We select one such theorem at random, in order to give an idea of how Runge's theorem can be applied. First we need a definition.

Let $f(z)$ be a function on the open unit disk $\mathbb{D}$, and let $\zeta \in \partial \mathbb{D}$. We define the **radial cluster set** of $f(z)$ at ζ to consist of all complex numbers w for which there is a sequence of radii r_j increasing to 1 such that $f(r_j \zeta) \rightarrow w$.

Theorem. *There is an analytic function $f(z)$ on the open unit disk $\mathbb{D}$ whose radial cluster set at any $\zeta \in \partial \mathbb{D}$ coincides with the entire complex plane.*

The proof construction goes as follows. Let $\{s_k\}$ be a sequence of positive numbers that increases to 1, let ε_k be a sequence of positive numbers that decreases to 0, and consider the circular arcs $\Gamma_k = \{s_k e^{i\theta} : \varepsilon_k \leq \theta \leq 2\pi\}$. Let S be a sequence of complex numbers that is dense in the complex plane, and let $\{w_k\}_{k=1}^{\infty}$ be an enumeration of S with repetitions, so that each number in S appears infinitely often in the sequence. Thus for any complex number w and for any $\zeta \in \partial \mathbb{D}$, there is a subsequence k_j of integers such that $s_{k_j} \zeta \in \Gamma_{k_j}$ and $w_{k_j} \rightarrow w$. Our strategy is to use Runge's theorem to approximate the function with constant value w_k on Γ_k by an analytic function $f(z)$. This is done by an induction argument, as follows. Let $f_1(z)$ be the constant function w_1 . Having chosen $f_{m-1}(z)$, observe that $\Gamma_m \cup \{|z| \leq s_{m-1}\}$ is a compact subset of the complex plane with connected

complement, so by Runge's theorem there is an analytic polynomial $f_m(z)$ such that $|f_m(z) - f_{m-1}(z)| < 1/2^m$ for $|z| \leq s_{m-1}$, while $|f_m(z) - w_m| < 1/2^m$ for $z \in \Gamma_m$. For $n \geq m$, $|f_n(z) - f_{n-1}(z)| < 1/2^n$ for $|z| \leq s_{m-1}$. Consequently, $\sum |f_n(z) - f_{n-1}(z)|$ converges uniformly for $|z| \leq s_{m-1}$, by the Weierstrass M -test. Hence

$$f(z) = \lim f_n(z) = f_1(z) + \sum_{n=2}^{\infty} [f_n(z) - f_{n-1}(z)]$$

is analytic on $\mathbb{D}$. From $f(z) = f_m(z) + \sum_{n=m+1}^{\infty} [f_n(z) - f_{n-1}(z)]$ we estimate $|f(z) - w_m|$ by

$$|f_m(z) - w_m| + \sum_{n=m+1}^{\infty} |f_n(z) - f_{n-1}(z)| \leq \frac{1}{2^m} + \sum_{n=m+1}^{\infty} \frac{1}{2^n} = \frac{1}{2^{m-1}},$$

for $z \in \Gamma_m$. Thus $\{f(s_{k_j}\zeta)\}$ has the same limit as $\{w_{k_j}\}$ as $j \rightarrow \infty$, which is w , and consequently, each complex number w is in the radial cluster set of $f(z)$ at ζ .

Exercises for XIII.1

1. Show that any analytic function $f(z)$ on a domain D can be approximated normally on D by a sequence of rational functions that are analytic on D .
2. Show that there is a sequence of polynomials $\{p_n(z)\}$ such that $p_n(z) \rightarrow 1$ if $\operatorname{Re} z > 0$, $p_n(z) \rightarrow 0$ if $\operatorname{Re} z = 0$, and $p_n(z) \rightarrow -1$ if $\operatorname{Re} z < 0$.
3. Let $\{z_j\}$ be a sequence of distinct points in a domain D that accumulates on ∂D , and let $\{w_j\}$ be a sequence of complex numbers. Show that there is an analytic function $f(z)$ on D such that $f(z_j) = w_j$ for all j . *Remark.* The sequence $\{z_j\}$ is called an **interpolating sequence** for analytic functions on D .
4. Let $\{z_k\}$ be a sequence of distinct points in a domain D that accumulates on ∂D . Let $\{m_k\}$ be a sequence of positive integers, and for each k , let $a_{k0}, \dots, a_{km_k}$ be complex numbers. Show that there is an analytic function $f(z)$ on D such that $f^{(j)}(z_k) = a_{kj}$ for $0 \leq j \leq m_k$ and for all k .
5. Let D be a domain in $\mathbb{C}$. Show that there is an analytic function $f(z)$ on D such that the cluster set of $f(z)$ at any point $\zeta \in \partial D$ includes the entire complex plane, that is, for any $\zeta \in \partial D$ and any complex number w , there is a sequence $\{z_j\}$ in D satisfying $z_j \rightarrow \zeta$ and $f(z_j) \rightarrow w$. *Hint.* Use Exercise 3.
6. Let $\{z_k\}$ be a sequence of distinct points in a domain D that accumulates at ∂D , and let E be a nonempty closed subset of the

extended complex plane $\mathbb{C}^*$. Show that there is an analytic function $f(z)$ on D such that E is the set of cluster values of $f(z)$ along the sequence $\{z_k\}$.

7. Show that there exist three analytic functions $f_1(z), f_2(z), f_3(z)$ on the open unit disk $\mathbb{D}$ such that the mapping $z \mapsto (f_1(z), f_2(z), f_3(z))$ embeds $\mathbb{D}$ as a closed submanifold of $\mathbb{C}^3$, that is, such that $|f_1(z)| + |f_2(z)| + |f_3(z)| \rightarrow +\infty$ as $z \rightarrow \partial\mathbb{D}$, and for each $z \in \mathbb{D}$ there is j such that $f'_j(z) \neq 0$. *Hint.* Take $f_3(z) = 1/(z-1)$. For $j \geq 1$, let E_j be the set of z satisfying $1 - 1/j \leq |z| \leq 1 - 1/(j+1)$ and $\varepsilon_j \leq \theta \leq 2\pi$, construct $f_1(z)$ to be approximately j on E_j for j odd, and construct $f_2(z)$ to be approximately j on E_j for j even.
8. Show that for any domain D in $\mathbb{C}$, there exist three analytic functions on D for which the mapping $z \mapsto (f_1(z), f_2(z), f_3(z))$ embeds D as a closed submanifold of $\mathbb{C}^3$. (See the preceding exercise.)
9. Suppose points $c_j \in \mathbb{D}$ and radii $r_j > 0$ are such that $\sum r_j < 1$ and such that the disks $D_j = \{|z - c_j| < r_j\}$ are disjoint subdisks of $\mathbb{D}$. Set $K = \{|z| \leq 1\} \setminus \bigcup D_j$, which is a compact set. By considering $\int_{|z|=1} (\bar{z} - f(z)) dz - \sum \int_{|z-c_j|=r_j} (\bar{z} - f(z)) dz$, show that

$$\sup_{z \in K} |\bar{z} - f(z)| \geq 1 - \sum r_j$$

for any rational function $f(z)$ with poles off K .

10. Let D be a bounded domain whose boundary Γ consists of a finite number of disjoint circles, and let $E = D \cup \Gamma$. Show that a function $f(z)$ on E can be approximated uniformly on E by rational functions with poles off E if and only if $f(z)$ is continuous on E and analytic on D . What functions on E can be uniformly approximated on E by polynomials in z ?
11. Let K be a compact subset of the extended complex plane $\mathbb{C}^*$, and let S be a subset of $\mathbb{C}^* \setminus K$. Let J be the union of K and the components of $\mathbb{C}^* \setminus K$ that do not contain points of S . (a) Show that J is compact and that $\partial J \subset \partial K$. (b) Show that if $\zeta \in J \setminus K$, then the function $f(z) = 1/(z - \zeta)$ is not uniformly approximable on K by rational functions with poles in S . (c) Show that a function $f(z)$ analytic in a neighborhood of K is uniformly approximable on K by rational functions with poles in S if and only if $f(z)$ extends to be analytic in a neighborhood of J . (Polynomials are regarded as rational functions with poles at ∞ .)
12. Let D be a bounded domain with smooth boundary Γ , and let K be a compact subset of D . Let $f(z)$ be a continuous function on $D \cup \Gamma$ that is analytic in D . Show that the Riemann sums $\sum [f(\zeta_j)/(\zeta_j -$

$z))(\zeta_j - \zeta_{j-1})$ for the Cauchy integral $\int_{\Gamma} f(\zeta)/(\zeta - z) d\zeta$ converge uniformly on K to $f(z)$ as the lengths of the arcs of Γ between successive points ζ_j of the partition tend to zero.

2. The Mittag-Leffler Theorem

Recall that if $f(z)$ is a meromorphic function with pole at z_0 , and the Laurent expansion of $f(z)$ at z_0 is given by $\sum_{k=-m}^{\infty} a_k(z - z_0)^k$, then the principal part $P(z)$ of $f(z)$ at z_0 is the sum of the terms with the negative powers, $P(z) = \sum_{k=-m}^{-1} a_k(z - z_0)^k$. Thus $P(z)$ is a polynomial in $1/(z - z_0)$, and $f(z) - P(z)$ is analytic at z_0 . The Mittag-Leffler theorem asserts that we can prescribe the poles and principal parts of a meromorphic function.

Theorem (Mittag-Leffler Theorem). *Let D be a domain in the complex plane. Let $\{z_k\}$ be a sequence of distinct points in D with no accumulation point in D , and let $P_k(z)$ be a polynomial in $1/(z - z_k)$. Then there is a meromorphic function $f(z)$ on D whose poles are the points z_k , such that $f(z) - P_k(z)$ is analytic at z_k .*

Let K_m be the set of $z \in D$ such that $|z| \leq m$ and the distance from z to ∂D is at least $1/m$. Then K_m is a compact subset of D , $K_m \subset K_{m+1}$, and each component of $\mathbb{C}^* \setminus K_m$ contains a point of $\mathbb{C}^* \setminus D$. Let $f_m(z)$ be the sum of the functions $P_k(z)$ for which $z_k \in K_{m+1} \setminus K_m$. There are only finitely many such functions $P_k(z)$. By Runge's theorem, there is a rational function $g_m(z)$ with poles in $\mathbb{C}^* \setminus D$ such that $|f_m(z) - g_m(z)| \leq 1/2^m$ for $z \in K_m$. Then $f(z) = \sum [f_m(z) - g_m(z)]$ converges uniformly on each compact subset of D , by the Weierstrass M -test. Since $\sum_{m=N+1}^{\infty} [f_m(z) - g_m(z)]$ is analytic on K_n for $N > n$, and $\sum_{m=1}^N [f_m(z) - g_m(z)]$ has poles at the points z_k that are in K_n , with the prescribed principal parts, $f(z)$ has the prescribed poles and principal parts in D .

In the case that D is the entire complex plane $\mathbb{C}$, we can take K_m to be the closed disk $\{|z| \leq m\}$. Then $f_m(z)$ is the sum of the functions $P_k(z)$ for which $m < |z_k| \leq m+1$, and we can take $g_m(z)$ to be a partial sum of the power series expansion of $f_m(z)$ about $z = 0$.

Exercise. Find a meromorphic function on the complex plane whose poles are simple poles at the positive integers with residues all equal to 1.

Solution. This corresponds to the principal part $P_k(z) = 1/(z - k)$ at $z_k = k$. Our first guess is to try $f(z) = \sum 1/(z - k)$, but the sum does not converge. The constant term in the power series expansion of $1/(z - k)$ about $z = 0$ is $-1/k$, so our second guess is to set $g_k(z) = -1/k$ and try

$$f(z) = \sum_{k=1}^{\infty} \left(\frac{1}{z-k} + \frac{1}{k} \right) = z \sum_{k=1}^{\infty} \frac{1}{k} \frac{1}{z-k}.$$

This series *does* converge, uniformly on bounded sets, by comparison with the series $\sum 1/k^2$. Indeed, if $|z| \leq R$ and $k > 2R$, then $|z - k| > k/2$, so the k th summand is bounded by $2/k^2$.

The technique of subtracting off terms of a power series expansion to guarantee convergence can be used to construct doubly periodic functions with double poles. Let ω_1 and ω_2 be two complex numbers that do not lie on the same line through the origin. We sketch the construction of the **Weierstrass P -function** $\mathcal{P}(z)$ associated with ω_1 and ω_2 . This is a doubly periodic meromorphic function on the complex plane, with periods ω_1 and ω_2 , such that the poles of $\mathcal{P}(z)$ are double poles at the lattice points $m\omega_1 + n\omega_2$, $-\infty < m, n < \infty$, and the principal part of $\mathcal{P}(z)$ at $m\omega_1 + n\omega_2$ is $1/(z - m\omega_1 - n\omega_2)^2$. Again our first guess for constructing such a function might be to sum these principal parts over all integers m and n , but the sum does not converge. If we subtract the constant term of the power series expansion at $z = 0$, as in the preceding example, we are led to a series that *does* converge,

$$\mathcal{P}(z) = \frac{1}{z^2} + \sum_{(m,n) \neq (0,0)} \left[\frac{1}{(z - m\omega_1 - n\omega_2)^2} - \frac{1}{(m\omega_1 + n\omega_2)^2} \right].$$

To prove that the series converges normally, we require an estimate for the number $N(k)$ of lattice points $m\omega_1 + n\omega_2$ in a fixed annulus $\{k < |z| \leq k+1\}$. We claim that $N(k) \leq ck$ for some constant c . To see this, choose $\delta > 0$ such that the distance between any two lattice points is at least 2δ . We consider the $N(k)$ disks of radius δ centered at lattice points in the annulus. Since none of the disks overlap, they cover an area of $\pi\delta^2 N(k)$. We may assume that $\delta < 1$, and then these disks are all contained in the enlarged annulus $\{k-1 < |z| < k+2\}$, which has area $6\pi k + 3\pi$. From $\pi\delta^2 N(k) \leq 6\pi k + 3\pi$ we obtain the asserted estimate $N(k) \leq ck$. If now $|z| \leq R$, and if $k > 2R$, then each summand with pole in the k th annulus is estimated easily by C/k^3 , where the constant C depends on R but not on k . Since $\sum N(k)C/k^3$ converges, the Weierstrass M -test shows that the series defining $\mathcal{P}(z)$ converges uniformly for $|z| \leq R$. Hence the series converges normally on $\mathbb{C}$ to a meromorphic function. Further, the series converges absolutely at each z , so that we may arrange the terms of the series in any order. For each fixed n the series

$$\sum_{m=-\infty}^{\infty} \frac{1}{(z - m\omega_1 - n\omega_2)^2}$$

converges to a periodic function of z of period ω_1 . Adding the appropriate constants and summing over n , we obtain $\mathcal{P}(z + \omega_1) = \mathcal{P}(z)$. Similarly, $\mathcal{P}(z + \omega_2) = \mathcal{P}(z)$, and hence $\mathcal{P}(z)$ is doubly periodic.

Let $f(z)$ be a meromorphic function on the complex plane, with poles at points z_k . A **partial fractions decomposition** for $f(z)$ is a normally converging series expansion $f(z) = \sum P_k(z)$, where $P_k(z)$ is a rational function whose only (finite) pole is at z_k . Thus $P_k(z)$ is the sum of the principal part of $f(z)$ at z_k and a polynomial. A partial fractions decomposition of $f(z)$ is never unique, as we can add any polynomial to one summand and subtract it from another.

Example. We show that $\pi^2/\sin^2(\pi z)$ has the partial fractions decomposition

$$(2.1) \quad \frac{\pi^2}{\sin^2(\pi z)} = \sum_{k=-\infty}^{\infty} \frac{1}{(z - k)^2}.$$

Set $f(z) = \sum 1/(z - k)^2$. If $|z| \leq R$, and if $|k| > 2R$, then $|z - k| > |k|/2$, and $1/|z - k|^2 < R/k^2$. By the Weierstrass M -test, the series converges uniformly for $|z| \leq R$. Thus $f(z)$ is a meromorphic function on the complex plane. The functions $f(z)$ and $\pi^2/\sin^2(\pi z)$ have the same poles and principal parts, so that $f(z) - \pi^2/\sin^2(\pi z)$ is entire. Evidently, $f(z + 1) = f(z)$, so that $f(z) - \pi^2/\sin^2(\pi z)$ is periodic with period 1. We may focus on the period strip consisting of $z = x + iy$ for which $0 \leq x \leq 1$. In this strip we have

$$\left| \frac{1}{(z - k)^2} \right| = \frac{1}{y^2 + (k - x)^2} \leq \frac{1}{(k - 1)^2}, \quad |k| \geq 2, \quad |y| \geq 1.$$

By the Weierstrass M -test, the series for $f(z)$ converges uniformly and $f(z)$ is bounded for $|y| \geq 1$ and $0 \leq x \leq 1$. Since each summand tends to 0 as $|y| \rightarrow \infty$, the uniform convergence implies that $f(x + iy) \rightarrow 0$ as $|y| \rightarrow \infty$, $0 \leq x \leq 1$. The identity $|\sin z|^2 = |\sin x|^2 + |\sinh y|^2$ shows that also $\pi^2/\sin^2(\pi z)$ is bounded for $|y| \geq 1$ and tends to 0 as $|y| \rightarrow \infty$. Thus $f(z) - \pi^2/\sin^2(\pi z)$ is bounded for $|y| \geq 1$, hence bounded on the vertical strip $\{0 \leq x \leq 1\}$, hence by periodicity bounded on the entire complex plane. By Liouville's theorem it is constant. Since both $f(z)$ and $\pi^2/\sin^2(\pi z)$ tend to 0 as $|y| \rightarrow \infty$, the constant is zero, and $f(z)$ coincides with $\pi^2/\sin^2(\pi z)$.

Example. From the partial fractions decomposition of $\pi^2/\sin^2(\pi z)$ we can obtain a partial fractions decomposition of $\pi \cot(\pi z)$ by integrating (2.1) term by term. This is justified on account of the normal convergence of